



FOREST POLICY BRIEF WEST PAPUA

Policy recommendations for the Green State Vision

Forest Policy Brief West Papua

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Cover photo: *A clear stream flowing through a lowland rainforest in West Papua.*

*Photo by Saul Mercado on Unsplash
West Papua, 2019*

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We are particularly indebted to those who have guarded and cared for the rainforest throughout millennia, whose names might have been lost to history but whose legacies live on in the extant forests and standing sago groves of West Papua.

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Positionality statement

Despite the goal of research to be as objective as possible, research involving the interpretation of the political goals and priorities of others on the basis of personal interactions or written resources is inherently highly subjective, as it is informed heavily by the social identities, values, and beliefs of the researchers and their sources. In recognition of this fact, we as the authors of this policy brief would like to take this space to acknowledge our social position and the ways in which this might affect our research.

Gijs Romijn is a white cis man who grew up in the Netherlands and has lived in several other European as well as South American countries. Svetlana Ehtee is a white cis woman who grew up in Serbia but who has lived in the English-speaking West for over a decade. Neither of us have ever visited West Papua. This position, especially given the history of Dutch occupation of West Papua and the sensitivity of the topic of West Papuan Independence, has possibly affected how we were perceived by the Papuans we spoke to. Despite our efforts to create a comfortable and open setting for our virtual conversations, it is likely that some interviewees retained a degree of caution and did not share all their thoughts freely. Language barriers have limited our ability to communicate directly and openly with Papuans, relying on interpreters or second languages instead, and linguistic and cultural differences between ourselves and interviewees might also have led to us interpreting responses differently than they were intended.

We gained our academic knowledge and scientific education in European and North American academic institutes. Therefore, our research tends to prioritize measurable results and concrete management measures rather than spiritual, relational, and process-based solutions, which would arguably be closer to customary West Papuan governance on a number of issues. While we consider ourselves critical of the modes of thinking about governance, politics, environment, knowledge, and humans' position in the world (including capitalism, human-nature dualism, centralized state governance, representative democracy, etc.) that are dominant in the Global North and although we have made active efforts to disregard some of the values we grew up surrounded by, our understanding of the world and consequently our research is undoubtedly still shaped by this background. We have made a conscious effort not to allow our personal views of the world to affect the contents of this research, although it is difficult (if not impossible) to disentangle our own views and beliefs from the attitudes presented in the Green State Vision (GSV).

In the end, we have tried to stay true to the goals in the GSV and our understanding of West Papuan cultures in this report and triangulated our interpretations whenever we could, but our distance from West Papuan reality on the ground does necessarily put us at risk of misinterpretation. We believe such risk, to a lesser or greater extent, is inherent to any form of representation, thus reinforcing the point that any government decision, including the interpretation and implementation of the recommendations presented in this policy brief, should be critically reviewed by a broad range of Papuans who live their everyday life in West Papua.

Reading guide

Explanations of **bolded** words can be found in the glossary in Annex I.

Footnotes are indicated by roman numerals^{IV} and used for additional notes, found on the bottom of each page.

Endnotes are indicated by decimal numbers¹² and used for references to the consulted literature, found in Works Cited.

Policy recommendations are indicated in **blue** throughout the text, and can be also found in the Compiled list.

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Key points

The political economy of West Papua is an essential factor in determining forest outcomes.

Secure land tenure for forest communities is essential for just and sustainable forest governance.

Promoting the full **inclusion of women**, and religious, tribal and ethnic **minorities in decision-making** at all levels of government is central to maintaining forest health.

Controlling international trade and monitoring corporate conduct should be a core responsibility of the West Papuan state.

The effects of policies regarding **road construction, industrial agriculture, and mining** are often more important to forest use than explicit forest use

Executive Summary

Forests are at the heart of life in West Papua. For many of the 250+ tribes in West Papua, the forest is a mother and a carer, the land of kin and ancestors, home and shelter, pantry and medicine cabinet. Over the course of more than 40,000 years the Indigenous human population of New Guinea has cultivated intimate connections and a delicate harmony with the various other beings that inhabit the island. As guardians of one of the world's major tropical forests with a unique geography and evolutionary history, Indigenous West Papuans have a relationship with forests that is of local and global importance.

Since West Papua's colonization by the Netherlands and the subsequent Indonesian occupation, Papuans have increasingly lost the ability to govern their forests. As the result of close and often militarized relationships between distant governments and the natural resource industry, West Papua's forests and all its inhabitants have come under threat. The interests of the Indonesian state and foreign corporations have created a **political economy and ecology** that is characterized by extractivist economics and paternalistic development discourse. Used to generate great short-term monetary wealth, the cultural, linguistic, and species richness of West Papua's forests have been paid little mind, and the self-determined needs of existing inhabitants largely neglected.

To reclaim the agency that West Papuans are owed and protect the reciprocal interspecies relationships they have nurtured, the ULMWP has presented an alternative vision of an independent West Papuan state that pursues an economy which serves the needs of all beings. The Green State Vision of West Papua envisions a society where governance is community-based, different species live together in harmony, and humans have the responsibility to be the guardians of local and global life. The West Papua Forest Policy Brief was drafted on the basis of an extensive in-depth study by Solon Law at the request of the ULMWP provisional government as a supplementary and subordinate document to the GSV, and builds on the foundations laid out by the Green State Vision to propose forest policy recommendations to support such a society.

Considering the various tradeoffs that exist between different forest uses, we find that the most beneficial balance for local livelihoods, customary practices, biodiversity, and carbon stocks is found in a landscape constituting an ecological mosaic that combines a high

degree of forest cover with agriculture based on agroforest home gardens and slash-and-burn practices with long uncultivated recovery times.

The outcomes of forest use are shaped to a great extent by national policy, although policy directly aimed at forest use has a relatively limited effect, which is a major reason why many national and international efforts to curb deforestation have failed to materialize. The power dynamics inherent to the political economy and ecology of a country affect how people and other actors on the ground relate to and use the forests they have access to, which ultimately determines forest outcomes for humans and other species alike. For a government to ensure sustainable and just forest use, it must address processes and policies across policy fields and governance mechanisms. For West Papua, this means that the national government should largely abstain from establishing specific forest management regulations and focus instead on defending community rights from outside interests and cultivating relationships of trust, respect, and reciprocity.

Trust is certainly required for the devolution of all rights and responsibilities of natural resource governance to communities, a policy approach that many governments are reluctant to fully implement. Community-based forest governance faces the major risk of elite capture, where a few individuals control and benefit from the use of forests, which contradicts their goal. However, when communities are supported by a solid regulatory system that builds on strong customary institutions, secure **land tenure**, community mapping, and inclusive methods of decision-making they can provide the greatest degree of self-determination and livelihood security for local humans, as well as ensure the conservation of species and ecosystem services.

Although foreign corporations and international trade have mostly brought harm to West Papua in the past, many Papuans remain hopeful about the economic opportunities that businesses might provide as well. There is reason to believe that if the negotiations between corporations and communities that are currently characterized by major power asymmetries take place on a truly level playing field there is potential for mutual gains. It should be the responsibility of the national government to safeguard the negotiation process and provide a policy environment that can ensure the extraction of and trade in forest products happens in a just way. We propose the government may create this space by supplementing secure land tenure with policies that: 1) provide clear standards for the establishment of Corporate-Community Partnerships, 2) enforce the fulfilment of the agreements, 3) ensure traceability of all exported forest products through a national forest registries system, and 4) strictly enforce trade regulations at border crossings.

Internationally the conservation potential of **market-based instruments (MBIs)** to generate income while protecting standing forests is receiving major attention and funding. We found that the commodification of standing forests through **ecotourism**, **bioprospecting**, and **payments for ecosystems (PES)** may indeed provide some potential for income generation while conserving species richness if it is protected from land grabbing and elite capture. However, the increased market orientation and cultural exchanges that come with it will have unpredictable and potentially far-reaching consequences for the diverse cultures of Papuans as well. We propose several scenarios that the government may adopt to balance these different priorities.

Extractive activities of forest products, such as logging, foraging and hunting, and cultivation through agroforestry also necessarily pose certain tradeoffs, as they disturb and manipulate the productivity of certain species within the forest but may provide important economic opportunities to local forest users. Conversions of forest land to alternative uses such as agriculture, mining, or road construction pose similar tradeoffs, although the impacts of conversion may be significantly more severe and permanent but also provide greater economic opportunities. Extraction of certain forest materials and

conversion of forest land for agricultural purposes has been a part of Papuan societies for millennia, and various natural disturbances are a normal and essential aspect of maintaining productive and biodiverse forests. The key to sustainable extraction and forest conversion is retaining the regenerative capacity of the various species and habitats within the forest. While modern science does not provide satisfactory answers to what the threshold for this should be, the intensity of **natural gap dynamics** and historic agricultural patterns based on **Traditional Ecological Knowledge** in West Papua may provide a suitable baseline that is likely to sustain the biodiversity currently found in West Papuan forests. The impacts of several policies in the fields of infrastructure planning and food security that may potentially increase harvest and conversion pressures, and the fragmentation of forests are also considered.

While community-level governance and interests form the basis of most of this policy brief, certain essential aspects of forests warrant a wider scope of governance, as the impacts of the activities of one community may have significant consequences for other communities within the country or across the globe as well. In particular, any governance of forests that relates to threatened species, carbon stocks, and watersheds needs to be protected at the community- as well as the landscape level. The most important policy tools available to the state in respect to these topics are designation of Indigenous and Community Conserved Areas, supporting inter-community networks and collaborations with scientists and NGO's.

All in all, forest policy in West Papua must pursue the leveling of power relations through the empowerment of local communities to allow customary laws, regulations, and practices to govern forests. The state must build relationships of mutual trust, respect, and responsibility, which requires that any policy, including those within this report, is only adopted and implemented after meaningful deliberation with those most affected.

A note on definitions

The definition of words used in policymaking shapes the information on which decision-making is based and affects the intended outcomes of laws and policies. An excellent example of this is the official definition of forests that the Indonesian government uses, which includes tree-based farms^{1–3}. Due to this interpretation, the conversion from biodiverse, carbon-rich, nourishing, old-growth rainforest to monocultures of oil palm or pulpwood trees is not reported as deforestation by the Ministry of Forestry, despite its extensive known effects on local livelihoods, species richness, ecosystem services, and carbon emissions.

A good working definition, we would argue, is one that integrates all the relevant considerations and objectives for the general population and policy makers, while remaining as concise and clear as possible³. We thus implore any government of West Papua to carefully consider the definitions of forest-related terms on which it would base its policies and to engage with forest-dependent Papuans to come up with functional and culturally appropriate definitions. We also provide template definitions that may be adapted and adopted by the West Papuan government:

Forest (definition)

Should include: primary forests, secondary forests, natural forest gaps, forest stands at different stages of succession, **long-fallow swidden agriculture**, native trees, mangrove and peat forests, history of a site.

Should exclude: arbitrary size and age requirements, tree plantations, monocultures, non-native trees.

Appropriate example definitions: ‘natural forest’⁴ or ‘intact forest landscape’⁵.

Proposed definition: An area of land dominated by a composition of a variety of native tree species of different ages that has not been completely cleared and regrown in recent history, including associated natural treeless ecosystems and temporary, small-scale anthropogenic and natural gaps.

Deforestation (definition)

Should include: permanent loss of trees and canopy cover, conversion to tree plantations, clear-cutting.

Should exclude: temporary forest clearing due to natural meteorological, geological, or biological phenomena and shifting subsistence agriculture, clearing of tree plantations.

Appropriate examples: follows from an appropriate definition of ‘forest’; FAO definition⁶.

Proposed definition: Permanent conversion of forest to another land use or the long-term reduction of tree canopy cover.

Forest degradation (definition)

Should include: large-scale/long-term alteration of tree composition, chemical/structural alteration of soils, decrease of canopy cover, homogenization of forest stands at the landscape scale, reduction of seed stocks and dispersers, reduction of genetic diversity, reduced availability of **NTFPs** due to human harvesting and invasive species.

Should exclude: small-scale natural and anthropogenic disturbances in line with pre-colonial rates.

Appropriate examples: varies per topic of interest (e.g. ‘reduction of carbon stocks’ for **REDD+**)

Proposed definition: Widespread or long-term reduction of the diversity of species, habitats, or environmental conditions and alteration of tree or soil composition in forests in excess of pre-colonial rates.

Introduction

On November 14th, 2021, the United Liberation Movement for West Papua (ULMWP) presented its Green State Vision at COP26 in Glasgow. The West Papua **Green State Vision (GSV)** expresses the ambition of West Papuan independence leaders to become the world's first green state and outlines the primary objectives and core pillars of their approach to 'make peace with nature in the 21st century'⁷. In this manner, the ULMWP hopes to provide guidance to the world in realizing the UN Environment Programme (UNEP) ambition of the same name⁸.

The West Papua Forest Policy Brief was drafted by Solon Law of the ULMWP provisional government as a supplementary and subordinate document to the GSV. This document aims to provide recommendations regarding the implementation of any policies relevant to forests in West Papua. The advice contained within this report may serve to inform West Papuan policy makers on forest policy approaches that are compatible with the goals and philosophy of the GSV and support the drafting and implementation of any relevant legislation upon West Papua's independence.

State of West Papuan forests

As of 2021, forests covered 83% (34.29 Million hectares) of West Papua'sⁱ land surface area⁹. West Papua's forests are of great importance to biodiversity and climate stability on local, regional, and global scales. The Australasian rainforest, of which 43% is located in Tanah Papua, is the third largest tropical rainforest in the world¹⁰ and the largest area of intact old-growth forest in Pacific Asia⁹. West Papua contains 3% of the world's tropical rainforests, 8.5% of its mangrove forests and some of the world's largest peat forests, which play an important role in climate resilience and mitigation⁹. The island of New Guinea is a biodiversity hotspot and the world's most floristically diverse island. Due to its relative isolation as an island as well as its great variations in elevation and topography, New Guinea is home to many unique species that cannot be found anywhere else on earth^{11,12}. West Papua is home to at least 7,616 species of plants, of which 58% are endemic to the island¹³. At least 4,624 species of vertebrate animals, 8% of the global total, have been found on New Guinea, including 4% of earth's known mammals and lizards and 10% of all described fish species¹⁴. Many species important to local communities and forest dynamics remain unknown to science to this day.

The importance of forests to West Papuans can hardly be overstated. Although other ecotypes, such as savannahs, alpine meadows, rivers, and coral reefs, are essential to the survival and identity of many Papuan peoples, it can be argued that forests are at the center of life in West Papua. Forest ecosystems and the various beings that call it their home are deeply entangled with every aspect of life for Indigenousⁱⁱ Papuans, who affectionately refer to it as their supermarket and medicine cabinet¹⁵. Aside from forests' role in meeting the sustenance, medical, and construction needs of many communities, Indigenous Papuans also experience an intimate sense of connectedness to forests and derive part of their identity from it, regarding the forest as their mother and certain species as their ancestors or their kin^{15,16}. Human interactions with other species and the greater ecological cycles of the forest in Papua are greatly shaped by the knowledge about their specific environments Papuan peoples have gathered

ⁱ Throughout this paper 'Tanah Papua' and 'West Papua' refer to all parts of the island of New Guinea and surrounding islands that are currently part of Indonesia and are claimed by the West Papua Green State; 'Papua' refers to aspects of New Guinea that exceed the boundaries of West Papua's current claim or predate the incorporation of the territory by Indonesia.

ⁱⁱ In this report 'Indigenous' refers to peoples that have developed a strong connection to specific lands through long-term settlement and intergenerational cultural transmission, rather than to its generic definition as 'inborn' or 'naturally existing' in a place. As such, in recognition of its use as a proper adjective and as a sign of respect, the term 'Indigenous' will be capitalized throughout this report.

and transmitted across many generations^{17,18}. This knowledge, commonly referred to as **Traditional Ecological Knowledge (TEK)** constitutes ways of life that can only be understood through living them, which speaks to the close entanglement of Indigenous knowledges, governance institutions, and spiritual practices found in West Papua¹⁹

While certain forest species such as sago palms and cassowaries enjoy nearly universal nutritional, cultural, and spiritual importance across West Papua, many of the species and habitats that humans in Papua rely on are highly specific to a specific environment and thus vary per tribe or community. Sites and forest types of high value to the physical and spiritual livelihoods of Papuans differ greatly throughout the island. The variety of valued sites, habitats, and peoples is fundamental to the species richness of New Guinea in many ways, which in turn is essential for local livelihoods. Conceptions of place and time are often relational, dynamic, and informed by a complex tangle of movements of species and spirits, historic events, and interspecies interactions²⁰. As such, the ethnic, linguistic, and ontological diversity of West Papua is closely linked to the biodiversity of its species, genetics, landscapes, etc. Any effort to conserve livelihoods and biodiversity in Papua needs to recognize, embrace, and support the very diversity of these human-environment relations, including their generational and gendered variations within any given community. However, it is noteworthy that primary forests are generally regarded as the most highly valued source of construction materials, food items, ornaments and rituals²¹, secondary forests and disturbed environments like roadsides, on the other hand, provide for significant hunting sites as well²².

Threats to West Papuan forests

West Papua's forests are often regarded as pristine environments with little human disturbance. However, humans have interacted with and depended on forests in New Guinea for tens of thousands of years. The island's biodiversity has not only persisted despite human disturbances, but also flourished due to a complex history of small and large human-made disturbances. While it is good to acknowledge that humans, when present, have significantly impacted West Papua's forests and their species' compositions for over 7000 years^{23,24}, current rates of habitat and species loss are unprecedented.

Drivers of forest loss (2001–2019)	Forest loss (ha)	Forest loss (%)
Industrial plantations: oil palm (97%) + pulpwood (3%)	208,223	28
Selective natural timber extraction	120,500	16
Inland water bodies	83,800	11
Roads + population centers	115,366	15
Fire	39,347	5
Mining	13,288	2
Other drivers	168,116	23
<i>Total forest loss</i>	<i>748,640</i>	<i>100</i>

Table 1: Major drivers of old-growth forest loss in Tanah Papua (2001–2019) – adapted from Gaveau et al. (2021)

Of the 2,479 West Papuan species listed on the International Union for Conservation of Nature's (IUCN) Red List website, 60 are critically endangered, 78 are endangered, and 128 are vulnerable. Many of these species are important to the medicinal²⁵ and nutritional²⁶ needs of West Papuan humans. Human pressures on these species include habitat loss, forest degradation, hunting for subsistence, commercial bushmeat, and pet-trade, invasive species, and harvest pressure²⁷.

In the first two decades of the 21st century, CIFOR⁹ and GFW² report a loss of forests in West Papua of 749kha (2%) and 660kha (1.99%), respectively. The main direct drivers of deforestation in this period were expansion of agribusiness in the form of plantations, selective logging, and infrastructure development (see table 1). However, it is important to recognize the interconnected nature of these drivers, such as the fact that logging often involves road construction, and that road construction increases the risk of conversion of forests to industrial plantations^{9,28}.

Forests in the West Papua Green State

In West Papua, centralized government programs aimed at economic development such as the transmigration program, the Trans-Papuan Highway, mega food-projects such as MIFEE and the Food Estate Program, and concessions for logging and plantation development underly much of the direct causes of deforestation^{20,29–35}. Thus, the policies that result in forest loss often are not forestry policies, but rather policies related to agriculture, transportation, land planning, and economy. Similarly, addressing the relation between business and state and the role of money in politics, be it through clientelist relations in elections³⁶ or from public-private collusion and corruption³⁵, has a greater impact on natural resource management than a legality-based approach. As such, the scope of this report is not limited to forestry but reflects an array of policy areas relevant to Papua's forests, a so-called informality-sensitive approach³⁶.

As this Policy Brief is informed by the Green State Vision, we aim to recommend a set of policies and regulations that can help to actualize its objectives for West Papuan forests. Consequently, we identified several guiding principles by which to assess the suitability of specific policy approaches to attain the GSV's objectives (see Annex 2). To guarantee the continued survival and wellbeing of all species in West Papua, forest landscapes should at least follow the 5 basic goals of ecosystem management: 1) protect sufficient habitat for populations of all native species, 2) represent all native ecosystems, 3) manage ecosystems at scale suitable for ecological processes, 4) plan long-term enough to allow for evolution, 5) allow for human use and occupancy³⁷. Based on these principles, we deduced that the ideal structure of forest landscapes in West Papua is one that constitutes a mosaic of different land uses and forest stands of different ages, with a high degree of connectivity between habitats and extensive stretches of old-growth primary forests (i.e. **Intact Forest Landscapes or IFLs**). Any policy that the West Papuan government implements should thus be compatible with attaining this vision.

The landscape that Indigenous Papuans have cultivated throughout history through their customary use and management of forests fits the description of such a diversified forest mosaic. Customary guardianship informed by the great variety of Indigenous systems of thought and practice found in West Papua is thus a powerful and appropriate foundation for forest policy in West Papua, anchored in the recognition and self-determination of Indigenous practices, philosophies, and protocols. Yet, any policy implemented in West Papua will also need to reconcile the novel reality of operating within a nation-state that operates inside a globalized economy, and with all the new cultural influences, market pressures, peoples, and knowledges that are present in contemporary West Papua. Tradeoffs do exist between these priorities, yet Papuan peoples and Indigenous customs have shown to be adaptive and capable of persisting throughout changing circumstances time and time again^{38–40}. Local and Indigenous communities may, for example, adopt modern monitoring and mapping techniques^{41,42}, engage in entrepreneurship independently or as **Community Forest Enterprises (CFEs)**^{43,44}, form partnerships with Corporations (**CCPs**), or build peaceful multi-ethnic communities through celebrations and communal protection of the environment^{38,40,45}. This policy brief attempts to recommend a set of policies that strike an appropriate balance between accommodating realities of the twenty-first century and allowing Papuan customary practices to flourish, with the ultimate hope that the West Papuan Green State may provide guidance to the world for navigating the unprecedented environmental challenges humanity currently faces.

Methods

To formulate suitable and effective forest policy recommendations for West Papua, we first conducted elaborate background research to gain a greater understanding of the state of forests in West Papua, the cultural context of customary Melanesian culture, the effects of different forest uses, and tradeoffs

between potential policy approaches. Our research has been informed by and has strived to answer the following questions:

1. *What is the current state of forests in West Papua, in terms of forest cover, forest types, biodiversity, and how does that compare to (pre-)historic conditions?*
2. *What industries and economic activities are the biggest drivers of deforestation, forest degradation, and forest land use change in West Papua?*
3. *What forest resources and environments do Papuans rely on most for their livelihoods and cultural practices?*
4. *Is the use of forest resources of any kind in West Papua compatible with the GSV?*
5. *Is forest land use change and/or deforestation due to other industries or infrastructure development compatible with the vision?*
6. *If they are, what kind of extraction of forest resources and forest land use change should be allowed?*
7. *What alternative (i.e. non-extractivist) methods for economic valorization of forests exist and are compatible with the GSV?*
8. *Should specific areas of forests be subject to higher standards of protection than others, and what kind of policy arrangement would best suit these areas according to the GSV?*
9. *What criteria, principles, standards, and exemptions should be applied in licensing economic activity in forest environments, and who should hold the power to grant such permits?*
10. *How should any duties and regulations imposed on forest users be monitored and enforced, and what role should customary authorities and provincial government play in this?*

To find the answers to these questions, the policy approaches and practices we studied were judged by their impacts on a number of factors including the security of physical and spiritual livelihoods of human forest communities, the economic opportunities available, the inclusion of marginalized groups of people, the chances of survival of more-than-human species, the provision of ecosystem services, the health and regenerative ability of forests, and the potential for carbon emissions and capture. The research that informed our answers to these questions, which forms the foundation of the recommendations and reasoning within this policy brief, can be found in the supplementary Supporting Research for West Papua Forest Policy file. In the subsequent writing of this policy brief we identified knowledge gaps that warranted further study, the findings of which are integrated into this brief but may not be included in the background research.

Our research is based mostly on academic literature and public research reports written and published by civil society actors, as well as virtual interviews and written exchanges with several academic researchers (see Annex 3). We recognize that there are significant limitations to this approach, including the fact that many aspects of Papuan society are not well documented in scholarly nor journalistic publications. Additionally, much of the information that does exist was produced by people and institutes from the Global North and who generally apply an analytical lens that was shaped and informed by ontologies that at best are foreign to customary Papuan thinking and at worst perpetuate and naturalize colonial modes of thinking and being. The authors of this report are, despite our recognition of this issue, prone to the risk of representing Papuan needs and interests in flawed or limited ways as well, which we expand on in our positionality statement. We have made efforts to speak to and interview West Papuans and West Papuan diaspora wherever possible throughout the writing of this report. We nonetheless urge the government of West Papua tasked with implementing forest policy to apply any recommendation within this report only after in-depth dialogue with a broad selection of forest-dependent Papuans.

Forest Policy Topics

1. Legal framework

The legal framework of a state constitutes the most fundamental aspect of any national legal system as it outlines the structure of government within the country. Since decision-making processes take place within the constraints of the legal framework, it affects the establishment of any and all laws and regulations within its respective jurisdiction. The duties, responsibilities, and rights that a state's government and citizens enjoy in relation to each other and to the non-human environment are defined by the country's juridical structure as well. Due to the range of policy areas affecting forests, it is futile to attempt to address forest policy without also providing recommendations regarding the legal framework that form the very foundation of the West Papuan state.

For the pursuit of establishing forest policy in line with the Green State Vision, West Papua's legal framework needs to be built upon, or at least be conducive to, customary Melanesian conceptions of governance and of interpersonal and interspecies relationships. West Papua's legal framework should also be suitable for dealing with the current reality on the ground and its colonial legacy, while creating possibilities for society to employ non-colonial modes of being and of governance. The nation's governance structure must give Papuan peoples the ability to exert agency and self-determination in the face of great power differentials, while simultaneously enabling the state to protect Papuan human and more-than-humanⁱⁱⁱ beings from abuse and empower marginalized groups of the West Papuan population. West Papua's laws must thus not be overly prescriptive for communities, but also form a solid basis for the formulation and enforcement of regulations at all levels of government.

1.1 Rights and responsibilities

The rights, responsibilities, and duties that all the actors in a society have to themselves, each other, and external actors are fundamental to shaping the relationship between them. While the concept of rights has been particularly important in addressing various injustices across the world, it has largely failed to protect the needs of human⁴⁸ and more-than-human beings⁴⁹. Commitments to human rights by state governments have arguably only made a difference in a few countries, where democratic mechanisms and civil society enjoy strong protection from a rule of law⁴⁸. Arguably, human rights are also inherently anthropocentric and speciesist, as they elevate the needs of humans above those of others^{49,50}. Rights of Nature are increasingly adopted by the Global North as a way to overcome their societies' **human-nature dualism**, and claim they are rooted in Indigenous thinking⁵¹. However, the very notion of rights relies on legal personhood⁵², which exists only in Western legal history and ultimately centers humanity as the basis for morality and legal standing⁵³.

Human-nature dualism and, consequently, the concept of inalienable, individually-held rights are foreign to Indigenous thinking and pre-colonial Melanesian culture. Adopting rights-based thinking as the basis for forest governance in West Papua may reinforce⁵⁴ and naturalize⁵¹ colonial modes of thinking that are ultimately detrimental to the social-legal-environmental relations that have maintained the biodiversity of Papuan forests up till now. Yet, rights-based thinking play a central role in international discourse, as evidenced by the Universal Declaration of Human Rights adopted by the UN in 1948⁵⁵. A rights-based approach to national and community governance should thus be understood as a strategic diplomatic tool in a colonized and globalized world, rather than an inherent property of Indigenous existence. Rights-based discourse may be employed by West Papuan governments to garner

ⁱⁱⁱ Although Indigenous Papuan understandings of interbeing relationships cannot be fully translated or expressed into a culturally distant settler language like English which contains certain connotations of time, space, and being that are incompatible with Papuan cosmologies, we use the phrase 'more-than-human' in this report to refer to all beings in Papua that are not part of the species *Homo sapiens*, in recognition of the agency, inherent value, and existence these beings possess independent of their relationship to humans^{46,47}.

international support and hold perpetrators of abuse accountable on the world stage, yet individual rights should not generally form the basis of law within the nation^{iv}.

The GSV, in line with customary Melanesian conceptions of reciprocity, centers guardianship rather than legal protection. As such, rather than assigning rights, a West Papuan Green State should focus on assigning responsibilities. This distinction can be understood in a legal context as a potential versus a mandatory mechanism of protection. Taken in this sense, ‘rights’ enable humans to demand protection for a right-holder on their behalf, whereas ‘responsibility’ mandates that protection is actually exercised regardless of legal intervention⁵⁶. It is important to note that what we mean by ‘rights’ is the duty of the government to protect the ability of the rightsholder in question to exert their agency in specific activities. When we therefore discuss ‘land rights’, we are not highlighting the privilege of any individual being entitled to certain land claims, but rather referring to what constitutes the responsibility of the national government to uphold customary claims to land and the communities’ ability to exercise their guardianship over it. In this context, abuse and environmental destruction are considered failure on the part of the responsible actors to uphold their duty to protect others, rather than a violation of others’ rights. In full cognizance of these distinctions, we do refer to certain collectively held rights in this report as this is the common verbiage in the sources we consulted when conducting our research.

Justice and governance should be based on responsibilities and duties of groups of individuals within an interdependent community of living beings, rather than on individual rights.

‘Rights’ in national policy should only be granted to collective entities and not be vested in individual humans.

Rights-based approaches to national and community governance should be understood as a strategic tool rather than an inherent property of existence and may only be embraced if the rights in question are held collectively.

1.2 Justice

The concept of criminal justice, which revolves around personal responsibility and restores balance in the world by punishing individual perpetrators, is part of the legacy of European colonialism. Criminal law’s focus on punishment offers the victims of trauma and environmental destruction little in terms of healing⁵⁷, and Papua New Guinea’s (PNG) experience with its inherited criminal justice system has demonstrated the shortcomings of such a system in a Melanesian context⁵⁸. In customary courts in Papua, justice is relational and spatial in nature and thus focuses on the restoration of harmony in place and social relationships⁵⁹. The notion of restorative justice seems to sit particularly well in the Papuan context because it focuses on healing and restoring harmony through storytelling and deliberation⁵⁸. Relational and restorative forms of justice often do contain an element of punishment, such as the notion of payback that exists in many Papuan cultures. Ultimately, however, restorative justice allows victims to heal and enables offenders to assume responsibility for their actions, in order to recreate a functional community⁶⁰. As such, restorative justice has the potential to be a decolonial and democratizing force⁵⁸ in West Papua, and offers possibilities for rebuilding trust and understanding between citizens, governments, and organizations⁵⁷.

^{iv} While we appreciate that individual political and civil rights, such as the freedom of expression, are important for good democratic governance in societies with a strong rule of law and could help bring in greater protections for individuals, we must emphasize that such rights carry Westernized modes of thinking and that, given the colonial influences in the region, the legislative apparatus of the Green State would need to account for such influences and adapt them to the Indigenous and local ways of living and governance. As weighing these concerns is beyond the scope of this study, more research on how to accommodate these potentially conflicting priorities is warranted.

The state should pursue, and implement mechanisms to achieve, relational and restorative forms of justice, rather than retributive justice, in response to (environmental) violations.

Legal procedures and punishments should center the healing of human and more-than-human victims, which requires facilitating dialogue and storytelling between victims and perpetrators, and in the context of forests may mean compensation for harms to human health and lost livelihoods as well as ecosystem restoration projects.

1.3 Governance structure

West Papua is a land that can be characterized by its diversity and great biocultural heritage⁶¹. Tanah Papua's ethnic, linguistic, and ontological diversity are closely linked to its biodiversity, as the Papuan forest landscape was formed through a history of diverse forest management approaches informed by cultural preferences⁶². To preserve the cultural integrity of the many Papuan peoples and the biodiversity of which they are the guardians, their great diversity of forest use and management approaches also needs to be embraced. While the formation of a West-Papuan nation-state necessitates a certain unification of identity that was unknown amongst New Guinean people in pre-colonial times, maintaining the diversity of customs and identities between different tribes, clans, and communities is just as important for the realization of the GSV. Considering the variety of customary authorities in West Papua and goals of the GSV, a **polycentric** system of governance, marked by strong community-level decision-making and customary legal institutions that sit nested within a national regulatory system, would arguably be well-suited for West Papua.

Polycentric governance is defined by 1) multiple governance centers; 2) an overarching system of rules; 3) spontaneous emergence of order and institutions from the exchange of ideas and methods. A polycentric system generally allows for greater diversity in decision-making and more satisfactory service provision than a centralized government can offer. In such a system, the national government provides a collective choice framework to formulate shared expectations and goals, and provides clear pathways for making collective choices and changing rules⁶³. Within a polycentric governance framework, communities have a great degree of self-determination when it comes to the governance of land and life within their territories but remain accountable to each other. Devolution of decision-making in this way is fundamental to **community-based environmental governance and natural resource management (CBNRM)** on a local level^{64,65}. However, informal networks of different actors and communities that establish shared regulations, inter-community learning, and mutual monitoring are also effective at landscape-level resource management through strengthening local voices and building trust between stakeholders^{63,66,67}. Potential weaknesses of polycentrism that need to be addressed within the Green State include: lacking social institutions, blame avoidance, inconsistent rule enforcement, persistent asymmetries in power and wealth, a lack of common ground between ideological camps, and problems that exceed the scale of that of the actors' interests^{68,69}.

A strong polycentric governance system relies to a great extent on local institutions. Institutions in Papuan Indigenous communities consist of the Traditional Ecological Knowledge held by that community, including customary forms of deliberation, decision-making, and law. Compliance with environmental regulations by community-members as well as by outsiders is greatly enhanced when such regulations directly respond to communities' concerns and are formulated and enforced at a local level⁷⁰⁻⁷⁴. As such, just and sustainable forest governance in West Papua requires the use of customary forest governance institutions such as *sasi*^{70,75-79} and deliberative spaces like *para-para*^{15,80}, and the recognition of customary laws and legal systems as a valid part of the national legal system^{60,67,81}. As such, West Papua should be a legally pluralistic country, where different approaches to law are nested within a national judicial system.

The state should employ a system of polycentric governance, where governance duties are executed and services are provided by multiple governance centers, and where the national government only provides an overarching system of rules and goals.

The state should impose different restrictions on harvest for subsistence and community use, and extraction for commercial use and exports.

The state must recognize its own duty to implement and enforce customary law in recognized customary territories.

The state must establish the legal authority of customary courts in national law, and nest customary courts at the community and tribe level inside the national judicial system.

1.4 *Division of responsibilities and duties*

In accordance with the goals of the GSV, best practices for community-based forestry⁶⁵, and the CBD's ecosystem approach⁸², the responsibility of forest governance should be decentralized and devolved to the lowest appropriate level, a principle known as subsidiarity. In general, the government should pursue a facilitative and refereeing role instead of an enforcement role, by aiming to prioritize local capacity building over centralizing governance tasks whenever possible. Bottom-up collaborations between communities and other customary authorities is preferable over centralized top-down approaches^{66,83}, so the West-Papuan central government may choose to promote and support inter-tribal organization in cases where such organizations do not organically emerge. Within customary territories, communities should make all decisions relating to forest and resource governance in a deliberative manner guided by customary norms and values. Regulations related to the use and extraction of timber and **non-timber forest resources (NTFPs)** should only be governed by customary communities, and local land planning regarding **land use and cover changes (LUCC)** and reserved areas should be done by those inhabiting the relevant territory.

In order to overcome the limitations of polycentric governance, protect communities from powerful external actors, and govern processes that extend beyond the boundaries of a single customary territory, the national government will also need to assume certain duties and responsibilities. Specifically, the state should have the responsibility to enforce legislation regarding corporate conduct and trade in forest resources³⁶, but refrain from enforcing regulations within customary territories unless support is specifically requested by the community. In addition, the state has an important role in outlining shared goals, promoting inclusive decision-making processes, establishing legal procedures, mandates, and jurisdictions, and providing additional checks and balances as well as mechanisms for conflict resolution.

Governance duties and arrangements related to forest lands should be divided based on the principle of subsidiarity.

Timber and NTFP harvesting activities and interventions relating to natural resource and land management in general should be governed by local communities.

The state must not impose legally binding restrictions or responsibilities on communities' subsistence use of forest resources.

The state must not enforce regulations within customary territories unless support is specifically requested by the community.

The state must not undertake any large-scale land zoning or planning on forest lands.

The state should have the responsibility to ensure inclusive decision-making processes at all levels of government.

The state may promote certain practices and organizational aspects relating to forest governance, such as the deliberative and transparent strengthening of local institutions and participation in national or intertribal associations of forest producers.

The state should prioritize customary methods of conflict resolution but provide third-party mediation and multi-stakeholder dialogues for when these are demonstrably exhausted.

The state should allow a national supreme court to hear cases that are appealed in customary and tribal courts and interpret the application of customary laws in doing so.

The state should support inter-tribal collaboration and organization with regard to watershed protection and species conservation by contributing financial and technical resources, as well as hosting inter-tribal events aimed at knowledge-sharing.

The state should create a national institute for forest health to monitor the status of threatened forest species and habitat connectivity at a landscape scale. This institute may relay observed and scientific information to affected communities and provide capacity building programs to support community management plans but must not have a legal mandate to impose specific restrictions or prohibitions.

Any national or regional advisory board that oversees landscape-level issues should be made up of representatives from different tribes, may operate as a working group of the customary council, and should be financially supported by NGOs, members, and/or taxes, but should operate independently from the national government.

1.5 *Funding government programs and services*

Although polycentric governance relies to a great extent on decentralized decision making and (often informal) local governance capacities, and thus warrants a relatively small central government, the state will still need funds for carrying out facilitative and monitoring tasks, for setting up capacity-building programs, and for providing certain services for social protection such as free healthcare, as outlined in the GSV. As a great source of material wealth and natural resources, forests present a potential source of state income. However, due to the multi-faceted importance of forest environments to Papuans, any effort to use forests to intensify income generation for the state will come with great tradeoffs for local livelihoods. Government-funded programs are unlikely to compensate for a profound loss of sense of self, community, and place, or for the nutritional, medicinal, and cultural value that communities derive from their forests, and should therefore not be funded at the expense of the well-being of forests.

Forests should therefore not be treated as a commodity by the state, and national policies should steer clear of industrializing forests and forest products to generate funds for government policies, as this will ultimately lead to the detriment of the very institutions that West Papuan society and Papuan identities revolve around. To some extent, tax revenue may be generated from commercial activities that rely on forest resources, in particular those intended for international trade, in the form of a sales tax or an excise. However, such taxes should not be excessively high, as this risks encouraging illegal trade in forest products and degrading any benefits received by producers and harvesters⁸⁴. Likewise, bilateral agreements with other countries' governments, such as the EU FLEGT program, might provide income opportunities for the West Papuan state^{85,86}. However, such legality-based programs risk impeding local forest benefits and increasing illegal workarounds, and should thus never be leading in West Papua's forestry policy. To avoid becoming overly reliant on forest-based revenue and pushing towards intensified forest use and extraction, West Papuan state income from commercial forest use should not constitute more than a small part of the total government budget and any effort to increase tax revenue should focus on other economic activities.

The state may levee moderate sales taxes on commercial activities that rely on the use of or extraction from forest environments and trade tariffs on exports.

The state may generate state income through bilateral agreements with other states only if they do not shape national legislation nor limit local decision-making or subsistence use of forests.

The state should prioritize non-forest sources of state income generation over forest-dependent sources of funding.

2. Land tenure

Secure land tenure is essential for communities to be able to exercise proper self-determination over their lives, to maintain their Indigenous identities, and to manage land and forests sustainably for future generations to enjoy^{18,43,71,87–91}. Concession-based tenure by corporate actors for the extraction of forest resources does not allow for customary guardianship of forests and has repeatedly failed to provide just and equitable access to and distribution of the potential benefits to communities. Fleeting or insecure individual or community-based land tenure can similarly lead to degradation of forests and the well-being of the populations that depend on a healthy forest environment, because individuals and companies are more compelled to extract what they can out of the forest before losing access to it, a process known as the ‘**tragedy of the commons**’^{64,92,93}. Elinor Ostrom’s theory of governance of **common-pool resources** avoids such environmental degradation by establishing a common property rights regime, which allows for communities to collectively own and govern the **commons** (in this case forest lands) they rely on⁶⁴. As a legislative blueprint that may support such land rights and help resist governmental and corporate overreach, we suggest a legislative structure similar to Article 27 of the 1917 Mexican Constitution^{94–97}.

Aside from establishing a firm base for Indigenous agency and guardianship of the forest, which are central goals of the GSV, devolution of land tenure and ownership to local communities is also associated with greatly improved livelihoods, retention of primary forests, and maintenance of forest carbon stocks. Loss of land rights harms the wellbeing of Papuans of both current and future generations, even when paired with increased income or well-designed community development programs^{33,98–100}. While global market forces can pose a threat to sustainable use of forests and local self-determination, CBNRM can defend community interests from governmental and corporate overreach, and has a good chance of promoting democratization within the community, reviving TEK and identity bonds tied to place, and magnifying economic autonomy^{17,101}. As such, **community-based forestry (CBF)**, and consequently secure community land tenure, is a central component of forest policy in West Papua.

2.1 Rights to what?

Secure land tenure can be established by protecting a specific bundle of rights, all of which are necessary for the sustained and effective governance of forests by communities. This bundle includes: 1) operational-level rights such as the right to access given land and the right to withdrawal of forest resources, 2) collective choice rights such as the right to management of internal use patterns and interventions, the right to exclusion of outsiders, and the right of alienation by selling or leasing the management or exclusion rights; 3) clear and long-term duration of aforementioned rights, and 4) rights to compensation and due process for cases where the aforementioned rights are revoked. The more firmly these rights are guaranteed, the more likely CBF is to succeed in its various goals¹⁰². However, to achieve a dynamic forest mosaic that can provide shelter and sustenance for all the beings within it, there is a need for preventing permanent and large-scale land-conversion, which suggests that land sales to powerful external actors, such as corporations or the state, should be limited. For this reason, land sales may be restricted to trade between members of the community and between customary communities⁹⁴. Aside from tenure security, related factors that significantly contribute to the property rights aspect of CBF are clear ownership, congruence between social and physical boundaries, rules and regulations, and to a lesser extent clear boundaries⁸⁸. To ensure inclusion and avoid elite capture, it is essential that the aforementioned rights are vested in and represented by a broad selection of the community, rather than an individual representative from the community.

The state should have the duty to guarantee the ability to govern ancestral lands to Indigenous communities by recognizing a bundle of indigenous land rights in the most

fundamental levels of West Papuan law, i.e. within the constitution, and for an indefinite duration of time.

The state should grant the ownership and usufruct rights of land and all resources contained within a territory to their respective customary communities, including minerals in the soil.

The bundle of land rights for Indigenous communities should include the rights to 1) land access; 2) resource extraction; 3) environmental management; 4) exclusion of outsiders; 5) alienation of aforementioned rights by sale (to the state or other communities) or temporary lease; 6) due process and compensation in case of the revocation of aforementioned rights.

The state should prohibit foreign corporations from owning forest land in West Papua. As such, the right to alienation of customary land should be restricted to sale within or between customary communities, and management and extraction rights may be leased out to corporations for a maximum of 50 years.

National law should apply a separability principle between the land and the buildings and plants on it: individuals and corporations may borrow or lease the rights to enjoy what the land produces from the land-owning community but may not own the land itself.

Contested areas and areas with overlapping land claims may not be exploited for commercial purposes and external parties must be prohibited from entering into agreements or contracts (including sale or lease) that apply to these areas.

Since land rights are owned by and assigned to the collective community and not individuals, any benefits derived from the exploitation of forests must be subject to equal benefit sharing.

2.2 Recognition process

Recognition of customary land claims is important for instilling a strong sense of tenure security, and for the prevention of conflicts around overlapping land claims. In certain cases, formalization of customary land claims is an essential part of establishing tenure security, resulting for example in greater maintenance of forest cover^{103,104}. However, formalization of tenure does not always support livelihood improvement and security, as it can result in greater administrative demands that disadvantage subsistence households¹⁰⁵, particularly by putting control over land at risk of corruption and capture by elites^{87,106}. Formalization of institutions, rules, and regulations related to land use can also present a tradeoff between tenure security and social cohesion, risking the exclusion of certain groups within the community⁸⁷. Formalizing land tenure also has the potential to weaken TEK and local institutions^{17,107}, and leave certain vulnerabilities ungoverned¹⁰⁸. Overlapping land rights in a system with formal land rights is even associated with more violent conflicts¹⁰⁹, of which the neighboring PNG is an exemplary case^{106,110}. In contrast, land conflicts between tribes in West Papua without formalized land rights remains rare¹⁵. Informal collaborative land and resource governance can often effectively regulate the management of natural resources and landscapes as well^{67,108}. In the end, there are some inherent tradeoffs to formalization of land tenure, as it creates ‘winners’ and ‘losers’ due to existing power inequities^{106,111}. The process of recognition, therefore, is more important than the question of whether customary land claims are to be formalized or not.

For land formalization processes to be just and lead to sustainable outcomes, it is necessary to incorporate equally distributed and transparent decision-making authority on both a community and a national scale¹⁰⁶. It is essential that any formalization process of land claims proactively engages with all ethnic group present within a territory, as well as with the women and youth within these groups of people⁴². Recognition and negotiation processes should also be based in genuine dialogue, and Papuan thinking about land¹⁵. Most importantly, it should not be overly bureaucratic, inaccessible, or costly in

terms of time or money^{15,87,110}. One feasible approach to increasing the recognition of customary territories regardless of formalization is the practice of participatory mapping, which may also play an important role in land governance planning as it allows for communities to have their most important spiritual sites, livelihood resources, and forest use zones to be recognized^{15,20,34,112}. Currently, government maps tend to represent the interests of powerful state and corporate actors by focusing on natural resources, and can be limiting in how they represent the dynamic lifeworld of the Papuans who live on these lands²⁰. Mapmaking is a process that is arguably political in nature as it necessarily makes compromises between creating tools that are operationalizable and representing the multidimensional richness of life, which makes it unlikely that one map can meet all the needs they aim to serve⁴². To overcome these limitations, technology can be applied to create dynamic and multilayered 3D maps that can include movement and relationships, and communities can be granted ownership and an active role in selecting the relevant layers to improve the decision-making that these maps are used for⁴².

Additional complications in the process of land recognition, and in community-based governance in general, include the matter of who is considered Indigenous and who should be included in customary decision-making, and the fact that many Papuans have been displaced and their ancestral lands have been occupied and at times destroyed. The matter of who is included in a community should be up to the community itself, although outcomes for environmental protection and local livelihoods within communities usually benefit from including migrants and non-Indigenous peoples in customary practices and decision-making processes, as they promote compliance and peaceful interactions^{38,45}. Leaving this vital decision to the community would also help avoid the issues facing the Sámi parliament in Finland where the power to decide who is and is not Sámi has been taken away from the Indigenous group, contrary to the right to self-determination granted by the UN Declaration on the Rights of Indigenous Peoples¹¹³. Creative peacebuilding through customary practices, environmental protection, and inter-ethnic and -religious dialogue may thus be promoted by the government as a means to reinstate harmony in Tanah Papua and invoke restorative justice. The redistribution of ancestral lands has the potential to be similarly restorative, although in cases where the forest has been destroyed or heavily degraded, or when communities have been removed for so long that they have lost important TEK, they must be supported in ways beyond simply reinstating their right to land. Specifically, this means providing technical and financial aid with reforestation projects or developing alternative source of livelihood and promoting mutual learning between different communities. In both cases the government must support these processes by providing conflict resolution mechanisms.

All land in West Papua should be recognized as belonging to customary communities and no forest land should be owned by the state unless it is purchased from the relevant community.

Recognition of ancestral lands should be established in law as the default position and not be dependent on formalization.

The validity of land rights should precede official documentation and formalization of the specific customary territories.

The state should avoid overly prescriptive paperwork and administrative requirements for the recognition procedures for customary territories.

The state should pursue the use of dynamic and multilayered mapping techniques.

The state should support the participatory mapping of customary territories with financial and technological capacity.

Communities should own and have free access to maps of their territory, and they should have an active role in choosing which layers are included whenever these maps are applied.

The state should promote the involvement of outsiders (i.e. non-Indigenous humans present in West Papua) in customary practices and rituals, as well as communal protection of the environment, and interethnic and intergenerational dialogue as tools for identity building, peacebuilding, and conflict resolution.

3. Inclusive decision-making & social protection

Besides secure land tenure, the way in which land-use decisions are made and through which institutions regulations are formulated is a central factor in determining the outcomes of community-based forest governance¹⁰². Inclusive decision-making and the protection of local interests are a result of how a community can act and present itself outwardly and how governance arrangements are made internally. To secure the ability for all communities to govern their land and exercise self-determination in a national and global context, they must have the option to operate free from the threat of violence and displacement. To do so, any military, policy or security force should be accountable to the public and not be under the control of those who are invested in the exploitation of natural resources. Sound democratic decision-making at local, regional, and national levels of government also requires preventing collusion and clientelist relations between businesses and the state, which are prevalent in West Papua under Indonesia's rule^{35,36}. Strengthening the legal basis for local and shared decision-making through pursuing polycentricity and establishing secure land tenure for communities are important components of addressing these issues from the ground up, but limiting the role of money in politics is similarly important in a representative democracy. In particular, regulations and campaigns need to be put in place to prevent vote buying and corporate capture of campaign finances³⁶.

To safeguard peoples' ability to maintain the integrity of their cultures and shape their identities, which is fundamental to their free expression and to the various diversities that constitute West Papua, it must be recognized and appreciated that the realities that peoples inhabit differ in fundamental ways, a concept also known as ontological pluralism^{42,46,114}. Governments operate within their own reality as well, which is shaped by the constituents it represents as much as by international diplomatic discourses such as the idea of sustainable development, but they must make sure that these specific realities do not come to dominate or erase local realities^{42,114,115}. To accommodate the different realities and ways of thinking of communities, the government must be cognizant of and allow for different interpretations of the language it employs in law and policy, and engage in continuous dialogue and social learning when working with communities. Communities should also be able to organize themselves in different ways. Some communities may be largely market-oriented or have the desire to exploit their forest resources as a formalized community-based businesses entity such as a **Community Forestry Enterprise (CFE)**^{43,44}, whereas others might lack the capacity or desire to formalize their business activities and remain unincorporated. Both forms should be recognized as valid entities that can engage in trade, partnerships, and forest governance equally.

In terms of the internal organization of community-based governance, the primary risk is elite capture, where a select group of people still make the decisions for the whole community and seize the majority of benefits derived from their labor and resources, leading to further marginalization of already vulnerable groups and increased corruption risk, and sometimes environmental degradation^{106,107,110,116–122}. When democratic institutions are functional within a community, CBNRM may lead to fair benefit sharing over time^{119,120}, but when unequal social structures persist they may worsen elite capture^{74,120,122}. Therefore, it is warranted that the national government takes a proactive role in the promotion of inclusive democratic decision-making processes within communities^{121,122}. However, state interventions should build upon and improve existing institutions, for example by increasing anonymity in the Noken system⁴².

The representation and inclusion of marginalized groups of people in decision-making processes are goals of the GSV in and of themselves. However, broad democratic representation and social protection within communities is of particular interest to forest policy as well since it affects forest governance outcomes and can stem environmental degradation. While customary community-based forest governance can address the marginalization of tribal and ethnic minorities in decision-making currently found in Papua, some tension may exist between customary practices and inclusive decision-making,

especially regarding the role of Papuan women. ‘Big man politics’ concentrates a lot of power in a few (mostly male) individuals, and women are often excluded from community-level deliberations in West Papua¹⁵. Similar patriarchal and hierarchical community structures in PNG¹¹⁰ and Ghana¹²³ have been detrimental to the potential social and environmental benefits of the great degree of land-ownership and decision-making power that communities enjoy there, respectively. Overcoming the exclusivist, elitist, and patriarchal dynamics in decision-making processes that support elite capture and marginalization of groups of people will thus require changes in both customary institutions as national policies. The state should be cautious not to impose its own governmental reality onto the realities of customary communities, yet may work with customary authorities to try to establish mechanisms that allow for more broad and inclusive engagement, for example by consulting a random selection of community members, or engaging with multiple informal community leaders from different social groups such as village mothers and youth leaders⁴².

The state must be cognizant of the divergent realities that exist within West Papua and avoid naturalizing its own reality and legal definitions by using legal language and engaging with communities in ways that allow for different interpretations.

The state must have the duty to engage with a broad cross-section of different social groups in a community when it collaborates with or seeks the free, prior, and informed consent of a community and may do so by consulting their respective informal leaders or by randomly selecting members of each group. This includes respecting the right of a community to withhold consent for any development that may affect their collective rights.

The state should allow the establishment of CFEs for the commercial exploitation of forest resources and the representation of the community in national and international markets.

The state should support communities in democratizing existing and forming novel management institutions through organizational capacity building and mediation.

Equal benefit sharing should constitute either direct payments to all adult community members individuals or collection in a community fund that is collectively managed through participatory budgeting or a democratically elected management board.

Payouts of benefits may not be appropriated or distributed by the national government.

The state must prohibit private payments to military, police, and security forces and require transparent budgeting for the military and the police force.

The state must restrict private donations to political campaigns and require transparent reporting on campaign finances.

The state must ban the practice of vote buying and support this prohibition with public awareness-raising campaigns and by monitoring candidates and corporate conduct.

3.1 *Women in forest policy*

Women in West Papua have an especially strong connection to Papuan forests, yet they remain excessively marginalized in Papuan society and sidelined in environmental decision-making^{15,124}. Papuan women know the forest surrounding their homes intimately due to their daily activities, and enjoy a special position as carers of the forest in the customary role of women as guardians and mothers of life^{15,125,126}. The exclusion of women from important decision-making in West Papua inevitably leads to increased deforestation and loss of customary control over land, as men are more compelled to sell land for short-term benefits than women^{15,127}. Aside from its benefits for forest use and protection, active inclusion in community decision-making and management activities also empowers women within their own households, increasing the likelihood of safe and just treatment¹²⁸. As such, the promotion of women’s role in environmental decision-making needs to be a central focal point of West Papuan forest policy for the realization of the GSV.

The state must guarantee the fair representation of women and tribal minorities in national and regional governments.

The state must establish equality between women and men with regards to the ability to hold and own land in the most fundamental levels of West Papuan law, notably the constitution.

The state should actively promote the inclusion of women in deliberative consensus-seeking discussions (e.g. *para-para*) within communities.

Women's participation in decision-making processes at the community level may take the form of women-only consultations or forums, alongside consultations that involve both men and women together.

CFEs and landowner companies must be overseen by a broad board of community members in which women and men are represented equally and hold comparably influential positions.

3.2 Traditional Ecological Knowledge

Indigenous and traditional methods of forest governance, knowledges, and ways of life are fundamental to the great remaining biodiversity and forest cover of West Papua, yet these approaches and the people promoting them are often marginalized in Indonesian contexts as well as in the global arena, where state-centered, technocratic management and science-based solutions constitute a global cognitive hegemony^{129,130}. However, TEK is now increasingly recognized by western scholars as an important element of sustainable forest and natural resource governance^{18,71,82,131–133}. In many cases, and especially when scientific knowledge is scarce, TEK can provide a better course of action for forest governance within a given context than any science-based approach could¹⁷. TEK does not only refer to information on the species and ecological processes within a given environment^{18,89}, but also involves institutions relating to harvesting and management practices of forest resources^{18,131,134–138}, as well as community rituals and decision-making processes^{19,46}. Various policies and societal pressures can form a threat to TEK, including changes in livelihood practices, religion, environment, education, and institutions, and the loss of transmission pathways and land rights. These changes might result from, amongst other things, marginalization, militarism, and market pressures¹⁷. It is important to note, however, that TEK is adaptive and can persist throughout changing circumstances, and may at times benefit to be supplemented with non-customary practices, ethics, and insights such as those derived from churches or Western science^{18,40,132}. For TEK's importance in sustainable forest use and in Papuan cultures, it should be actively protected and promoted as a means of forest governance. Specifically, TEK should inform environmental regulations and harvest restrictions (e.g. *sasi* or *teter*)^{21,40,70,76–78}, land use planning within customary territories^{131,137}, and protection of threatened species (e.g. through taboos or rituals)^{21,40,46,71,132}.

The state should encourage inter-community organization in forest-producer associations and networks.

The state must legislate on the customary communities' rights to intellectual property, including traditional knowledge with regards to the use and processing of forest resources.

The state must avoid human-nature dualism in government discourse, policy, and law.

The state must avoid standardization of education and include TEK and local languages in the public educational curriculum.

The state should invest in capacity building for Indigenous artisanal practices and skills.

4. Corporate-Community Partnerships

While the track record of the relationship between foreign companies and Indigenous Papuans is full of violence and abuse, some Papuans also appreciate the economic opportunities that certain companies

bring^{15,139,140}. At least in theory, allowing companies to extract resources from West Papuan forests could attract investment in the well-being of Papuan communities and provide financial and technical resources as well as market access, the lack of which could limit the ability of forest communities to monetize their forest resources⁶⁵. Due to the collusion with and corruption of national and local governments plaguing West Papua under Indonesian control³⁵ and false promises of benefit-sharing and village development^{15,141}, companies are currently mostly harmful to Papuan forests and communities. Although there are tradeoffs between the availability of NTFPs for the subsistence use of human and more-than-human forest-dwellers and the potential economic benefits of commercial extraction of resources from forest land, forest communities can presumably benefit from forming **Corporate-Community Partnerships** (CCPs) if they have a fair and equal bargaining position and are able to engage with or dismiss potential partners freely¹¹⁰. Secure land tenure and inclusive decision-making are fundamental to securing such a bargaining position, but even with those in place there are pervasive structural, institutional, and financial inequities that can interfere with a fair negotiation process. The national government, therefore, must actively ensure that companies and communities establish fair agreements and enforce the terms of these agreements when necessary. CCPs may be controlled by a form of license that in this report we refer to as a partnership permit, approving that the requirements for a fair agreement have been followed. Experiences from PNG, Ghana, and Canada may provide guidance on what should be required for the approval of such a partnership permit¹¹⁰.

4.1 Procedure of establishment

In terms of procedure, the state should formulate clear procedural standards with regards to developing Company-Community Partnership agreements. It is necessary that agreements are established with individual communities after broad, free, and meaningful engagement between the potential partners. Currently, forest land in West Papua is often conceded by communities because the company already has a permit from the national government. In a Green State this process should be reversed: a community should give its consent to the terms of a project before a national license may be granted. Two options that could provide this type of process are 1) a community-led application procedure, or 2) a state-led tender procedure. A community-led procedure would require community members to initiate the application for a partnership permit, which would mean negotiations between corporation and community would be truly free and predate any national government involvement. A potential benefit of such direct negotiations is that it would prevent excessive collusion between the state and corporations. However, a community-led application process would require significant effort on the part of a community, which could discourage corporate involvement in the exploitation of forests altogether or serve as evidence that a community is truly motivated to work with the company when applications are filed, both of which could result in sustained forest cover and just forest exploitation. As of now, we are not aware of any country who has successfully implemented such a system, although this could be attributed to the general reluctance of state governments to fully devolve decision-making power to communities^{65,87}.

Alternatively, to ease the administrative burden on communities that want to enter into a CCP, the state might take a more active role in the process. Ghana has had some success with state-led tenders, in which a government employee first visits the community to consult them about their desires and conditions for agreeing to a CCP. The communities' wishes as well as a description of the land and other relevant technical details are then included in a tender offer to which companies may submit a bid that includes a plan for community engagement, benefit sharing, and investments in village development. Bids are evaluated by a government committee on the basis of non-financial aspects and awarded to the company that proposes the most beneficial social provisions, after which the company is still required to negotiate a **Social Responsibility Agreement (SRA)** including a code of conduct and a statement of social obligations with the community. This approach is lauded for a number of aspects, including the legal clarity of minimum standards for companies, the transparency and livelihood focus of tender procedures, the direct nature of the negotiations between both partners with a refereeing role for the

government, and the simplicity of the format¹¹⁰, but has failed to guarantee equal benefit sharing as companies negotiate with tribal leaders alone and most community members lack legal knowledge on their right to be included^{116,123}. An additional downside of this system might be that it requires significant labor resources from the national government, which in turn could lead to pushing for CCPs more forcefully, thereby increasing the risk of forest degradation and encroachment of communities' self-determination. In either case, it is of utmost importance that broad community participation and the right to self-determination are guaranteed, which includes the right to refuse to enter a CCP or to engage in negotiations with the state or a company³⁵.

Communities may enter into a partnership with an external (foreign or domestic) company (CCP), either as a CFE or as a community without a registered business license.

Consent-seeking procedures and Social Responsibility Agreements should be bottom-up rather than top-down: a community needs to consent before any licensing may be granted.

The state may employ a community-led or state-led application procedure for the establishment of CCPs.

Agreements within a CCP should be made individually with each community involved.

CCPs may not exploit resources located on land that is disputed by neighboring tribes or clans.

4.2 Contents of CCP agreements

CCP agreements may pertain to the exploitation of timber, NTFPs, **ecotourism**, **payment for ecosystem services (PES)**, or any other type of arrangement where an external company assumes the ability to manage or use part of a community's territory. Depending on the scope of the agreement, it must make sure to be in accordance with the relevant regulations pertaining to the topics discussed in article 5-11 of this report. Regardless of the scope, however, the CCP application and SRA should be accessible to and contain proof of strong support and awareness from a broad cross-section of the community¹⁴². The SRA should contain a code of conduct, a statement of social obligations accounting for important places, dates, and practices within the territory, and a description of conflict resolution mechanisms and terms of termination. The technical aspects of the agreement should also be included in the form of a community forest management plan that contains at least an overview of the territory made through participatory mapping, plans for which activities are executed where, which areas will be reserved for wildlife conservation and spiritual reasons, and which NTFPs will be actively managed.

The state must be responsible for safeguarding the just treatment and consent of communities in CCPs and may require the involved company to provide documentation to this end.

All documentation and reporting that the state requires must be submittable in multiple forms (e.g. written, video, audio, etc.) to ensure their accessibility for non-literate people.

Companies applying for a CCP must have the duty to provide proof that broad awareness and support of the project plans exists among both male and female community members.

CCP agreements should include: 1) a community-made map of the territory; 2) a statement of social obligations specifying the culturally specific sites, practices, and dates to be respected; 3) a benefit-sharing and redistribution plan; 4) a forest management plan; 5) a description of the division of responsibilities; 5) a plan for the continued engagement between the partners in a way that is appropriate to the community's customs and capabilities; 6) a consideration of the long-term future (at least 50 years); 7) a clear description of mechanisms for conflict resolution; 8) an explicit agreement on lease length and terms of termination.

CCP forest management plans for extractive activities should include: 1) a protected area that covers a significant portion of the primary forest, connects to forests on neighboring territories, and represents all ecosystems within the territory; 2) a harvesting plan according to **Reduced Impact Logging (RIL)** guidelines; 3) a full stand inventory; 4) a management intervention plan that includes a strategy to minimize negative impacts on NTFPs including an indication of expected social and ecological benefits; 5) a commitment to the registration of all extracted resources in forest product registries.

4.3 *Reporting, monitoring, and enforcement*

Community members have the most intimate knowledge of the forest and of the conditions of the CCP agreement and are able to conduct monitoring of activities within their territory as part of their daily activities. Monitoring and enforcement of the terms of agreements will thus first be the responsibility of the community. However, it should be the ultimate responsibility of the national government that agreements are fulfilled and to hold companies accountable for unfulfilled promises, as violators could easily leave the jurisdiction of the community otherwise. Communities should thus be able to complain to a monitoring body in case companies or groups of individuals illegally intrude on their land to extract resources or violate the terms of their agreement. The state may also impose a duty on companies to report periodically (annually/ biennially) on harvesting rates and company-community relations in order to monitor corporate conduct on the ground¹¹⁰. Additional checks and balances could be provided in the form of an ombudsperson, which has played an important role in PNG in terms of keeping the national government accountable for their encroachment on landowner rights in their effort to open up forest “development” to foreign companies^{143,144}.

Companies in a CCP should have the duty to report biannually on activities relating to company-community relations and on stand stocks of timber and NTFPs.

The state should create the position of an environmental ombudsperson that operates independently from the government and is responsible for ensuring national and local policy is in line with securing inclusive and community-level forest and natural resource governance.

5. Forest product registries

Flow-based arrangements such as ecolabels are often preferred by corporations and governments over territory-based governance because they facilitate trade, even if they do not perform as well in terms of environmental protection and social and democratic inclusion^{145,146}. However, governance arrangements arguably are most effective when they work in tandem. Aside from the structural power differences that underly the global abuse of local and Indigenous communities, the lack of transparency in supply chains is also a factor in the continued consumption of products that cause environmental destruction and human rights abuses. To be able to hold companies accountable across country borders and enforce fair prices and just treatment for forest communities across the globe, increasing traceability is beneficial. Mandatory traceability of exported forest products could thus allow West Papuans to engage in the global market while enjoying greater protection from corporate abuse, corruption, and organized crime^{147–149}.

When forest products can be traced back to the community they originate from, the national government can also monitor and regulate extraction of forest products without the need to invest greatly in the capacity of monitoring and auditing staff, and without violating the territorial integrity of communities. Data on species occurrences and harvesting rates may also inform environmental governance decisions and assist with the conservation of threatened species on a landscape scale, which would benefit West Papuan communities who rely on forest species and environments in the long-term. This might be especially valuable considering scientific data on most species in West Papua is lacking^{132,150}. The most feasible (i.e. technically accessible and low-cost) way to implement a system of traceability of forest products is to maintain registers for all logged timber, harvested NTFPs, and hunted game at the community and national level¹³². In such a register, community enterprises and CCPs must record all resources that are extracted for commercial purposes in the appropriate community register, and processing facilities such as sawmills must record the origin of their raw materials and into what they are converted¹⁵¹. These registers should be periodically reported to the national government, which is responsible for maintaining an open-access national registry that integrates all community-level data. If and when sufficient financial resources are available this traceability system would benefit from being digitized¹⁴⁸, which allows for real-time cross-referencing but require significant capacity building, integrating blockchain technology^{152,153}, which reduces the risk for fraud and corruption, and executing audits using genetic or molecular fingerprints^{154,155}.

The state should create a national forest products registry system for monitoring the flows of all timber, animal, and plant products extracted from forests intended for commercial trade.

Communities that wish to sell their forest products to an international market should be required to maintain and report on a register for each class of export product.

Sawmills should be required to maintain and report a register that notes the origins of the timber they receive and to what products each of these converted.

Traders should be required to be able to present a full registry of the origins for each product they are exporting.

Border and customs agents should have the duty to copy and enter the relevant export data into a national registry.

The state should allocate the responsibility of cross-referencing the reported registries to an internal department.

The state should establish and delegate funds to a separate investigative department, or hire a third-party, to conduct occasional audits at all steps within the forest resource export chains.

6. Trade and export

Since international trade is one of the biggest drivers of deforestation and forest degradation¹⁵⁶, and because West Papua has been part of the largest and most destructive illegal trade route in the world in the past¹⁴¹, regulating trade must be an important part of the responsibilities of the West Papua national government, especially as it relates to sustainable forest policy. Monitoring and enforcing trade at border crossings is an effective and cost-efficient approach to regulate the extraction of forest products, as previous crackdowns have demonstrated¹⁵⁷. To do so effectively, West Papua must have a solid customs system and collaborate with PNG to coordinate legislation and border control to avoid leakage of forest products across the border.

Regimes focusing on the legality of harvesting of forest products have consistently failed to reduce deforestation or improve livelihood opportunities in West Papua^{147,158,159}, so the government should pursue a policy of traceability instead, building a system of export licenses around its national forest product register. Export licenses should only be granted to parties that can demonstrate from which communities their products are sourced by presenting a full registry of the origins for each product they are exporting. These registries should be copied and entered into a national registry and cross-referenced with the registers as reported by the communities themselves by an internal department. Additional important trade regulations are restrictions on the export of raw logs and wildlife for the pet-trade. Intercepted shipments without a proper export license and findings of fraudulent registers should be punished firmly, and efforts should be made to return confiscated goods and charged fines to the community from whom they were stolen. The perpetrating company and its owners should be denied access to future export licenses and company executives higher up the chain must be held accountable.

The international borders of West Papua, on land and at sea, must be a central aspect of monitoring and enforcement of forest use and resource extraction by the national government.

The state must make significant investments in setting up a strong and transparent border customs system.

The state should pursue full transparency and traceability of forest product supply chains to facilitate ethical trade, and to limit corruption and environmental abuses.

Export licenses for forest products should only be granted to parties that can present evidence of where the transported products were sourced at a community-level.

The state should impose fines on illegal exports that are proportional to the captured cargo's value, for which the minimum fee exceeds at least the total value of the confiscated products.

Confiscated goods and charged fines should be returned to the community from whom the goods were stolen if possible or used by a government agency to support communities active in forest product trade.

Companies that violate trade regulations and their owners should be denied access to future export licenses and efforts should be made to prosecute the executives higher up the chain.

The state should also collaborate with the PNG government to coordinate legislation and oversee the land border to avoid leakage of forest products across the border.

The state should ban the export of uncut logs.

The state may impose an export tax on all forest products transported across the border.

7. Market-Based Instruments

Various Market-Based Instruments (MBIs), notably ecotourism, bioprospecting, payment for ecosystem services, and **REDD(+)**, have been designed to provide a source of income for forest communities without relying on the extraction of forest resources. MBIs are increasingly promoted internationally to protect forests while benefiting local communities that work according to free-market principles. The underlying reasoning of MBIs, also known as stakeholder theory¹⁶⁰, is that when the financial benefits of protecting the environment outweigh the opportunity costs of foregoing forest extraction or land conversion, people will not be inclined to act in ways that cause deforestation or environmental degradation. Although there might be merit to this reasoning in certain contexts, it can also be argued that the rationale behind MBIs is based on the same economic thinking that has been the driver of environmental degradation worldwide¹⁶¹, and that their functioning relies on low prices that result from power imbalances between the Global North and South¹¹¹. While this in and of itself does not disqualify MBIs for strategic use in the GSV, their application will require careful consideration, especially as their influence on culture can be pervasive.

REDD+ and other PES schemes form a particular risk of bringing about land conflicts and ‘green grabbing’, while failing to provide real financial benefits without significant financial contributions from government funds^{111,162}. Additionally, REDD+ is considered a risky and misleading climate mitigation strategy as it justifies emissions elsewhere on earth, yet it can be effective way of receiving money from foreign governments^{111,161,163–165}. Ecotourism, on the other hand, is the only MBI that is reliably able to offer additional income to communities, and existing projects in West Papua are considered successful, but it is associated with profound cultural changes as well as road construction and excessive carbon emissions from air travel^{131,161,166,167}. Bioprospecting involves the trade in genetic resources and traditional practices, which poses the most minimal risks to Papuan cultures and forest landscapes, but it is also a deeply unreliable source of income for communities. Bioprospecting is difficult to control due to the easy access to genetic resources in neighboring countries, and good enforcement of regulations relies heavily on the legislation in producer countries, despite the fact that the Nagoya protocol established international guidelines^{107,168–171}.

Each MBI has its particular strengths and weaknesses, but one aspect that is shared between all is that they rely on a certain commodification of nature¹⁶¹. In the process, it risks turning local intrinsic values, which for Indigenous Papuans are rooted in deep spiritual and cultural connections to their space and sense of self, into economic values, which can be exchanged on a global market¹⁷². Arguably, MBIs form a compromise for continued existence within a globalized and colonized world, where the conservation of forests can be secured by assimilating to the cultures of the Global North. Commodification of TEK, for example through bioprospecting or the sale of cultural artifacts, can simultaneously be a threat and a conservation measure¹⁷. While in some cases such commodification could still lead to positive outcomes for West Papua’s inhabitants, it does weaken the tested, specialized ecological knowledge and cultural institutions that have made West Papuan diverse culture and landscape what it is today. The economic incentives that PES offers for control over land also pose an additional risk for the power dynamics within the country and may pose a threat to marginalized peoples and customary territories. The West Papuan government must thus weigh its different priorities in order to choose which path to pursue in relation to MBIs, which is why we outline 3 different scenarios in this report: 1) restriction, 2) passive use, and 3) active promotion.

The state must simplify legislation, regulatory language, administrative procedures, and communication with communities regarding MBIs as much as possible.

The state should facilitate the engagement of communities with any MBIs by means of legal capacity building and offer third-party mediation where necessary.

The state should invest in social learning for its employees to achieve a clearer understanding and better definitions of local customary and legal institutions, knowledges, and needs.

7.1 ABS & Nagoya protocol

Correctly implemented, the principles of Access to Genetic Resources and fair and equitable Benefit Sharing (**ABS**) from the Nagoya protocol could help generate additional income for communities while protecting West Papua's biodiversity. Yet, challenges which cannot be overcome through national policy alone remain. Regardless of which scenario is adopted, the Green State should become a signatory to the Nagoya Protocol and implement appropriate bioprospecting regulations to protect communities' intellectual property rights and genetic resources from biopiracy. Foreign corporations have excluded Indigenous people from using their own genetic resources and customary practices through foreign patents^{173,174}. Additionally, the risk of misappropriation of resources in a Traditional Knowledge Digital Library (TKDL) for the purposes of "biocapitalism"¹⁷⁵ or as a means to enhance national and international interests without appropriate regard for the relevant local communities¹¹⁵ should be avoided by making certain that a solid intellectual property system requiring reporting on **Prior Informed Consent (PIC)** and **Mutually Agreed Terms (MAT)** is in place¹⁶⁸.

The state should become party to the Nagoya protocol and implement the required legislation on a national level.

The state must be responsible for protecting the ability of Papuans to use and trade all customary crops and genetic resources from encroachment by foreign patents.

The state must require proper reporting on PIC and MAT when granting patents on natural and genetic resources, and promote this practice internationally.

7.2 MBI scenarios

Scenario 1: Restriction

Strengths - Based on customary practices - Limits risk of land grabbing & corporate abuse - Low-cost monitoring and enforcement	Weaknesses - Limits income generation & livelihood diversification - Slightly limits self-determination of communities
Opportunities - Retention of culture & identity, including TEK - Forest protection through customary governance	Threats (Illicit) Extraction & trade of forest resources becomes more likely

A restrictive scenario would require the prohibition of certain or all MBIs. While this approach could limit the economic opportunities available to West Papuans that rely on forests for their livelihoods, it would likely retain a great variety of localized cultural identities and approaches to forest use and governance. For those who are content with customary ways of living, relying on subsistence-based farming, foraging, and local trade would result in a great degree of forest protection. However, banning all MBIs does limit the variety of livelihood options and poses the risk that changing circumstances like increased market pressures or population growth will lead to forest degradation. Allowing ecotourism and bioprospecting while banning international PES and REDD+ schemes from operating in West Papua, could create realistic opportunities for income provision without allowing an excessive amount of green grabbing and shifting power relations.

Ban international PES schemes, including REDD+.

Allow communities to engage in community-based ecotourism projects and bioprospecting.

Prohibit foreign and corporate investments in ecotourism business and infrastructure.

Scenario 2: passive use

Strengths Greatest self-determination for communities	Weaknesses Administrative burden on communities
Opportunities Organic dissemination & selection of beneficial arrangements	Threats Increased inequality between communities

A passive use scenario would focus on maximizing the self-determination of communities. The national government does not actively prohibit nor promote the use of MBIs, but allows communities to engage in ecotourism, bioprospecting, and PES schemes on their own accord, or through CCPs. The administrative and organizational burden on communities might initially limit the frequency in which communities engage with MBIs, but projects that are perceived as beneficial by the community could spread organically through the country. Not having an active policy on the application of MBIs might leave smaller communities that are not able to access such opportunities to the same extent vulnerable to further marginalization. To secure the inclusion of customary practices and protect poorer communities from exploitation by foreign actors, a few basic policies should be adopted by the national government.

Allow communities to engage in ecotourism, PES schemes (incl. REDD+), and bioprospecting.

Promote the use of inclusive and pro-poor policies in PES schemes.

Include traditional land uses (e.g. **swidden fallows** and agroforests with maintained canopies) in REDD+ and other PES schemes.

Require the establishment of a CCP for all PES schemes and for external parties partaking in ecotourism, including hotel owners, tour operators, and travel agents.

Pursue the formation of a genetic resources 'cartel' with neighboring countries to protect genetic resources and customary practices from a 'race to the bottom'.

Scenario 3: active promotion

Strengths Greatest income generation for state	Weaknesses Costly administrative, monitoring & enforcement burden on state
Opportunities - Livelihood diversification for households - Forest protection through stakeholder theory	Threats - Degradation of culture & identity, including TEK - Homogenization of landscapes - Increased inequality between & within communities

Actively promoting the use of MBIs in West Papua will allow most monetary resources to flow into the country, but also put local communities at the greatest risk of being affected by the coercive forces of the state and market forces. A proactive approach to MBIs will likely bring about unintended consequences within communities and on a national level, especially through the increase of market-

based thinking and homogenization of landscapes and governance approaches. If this money is invested properly into forest conservation efforts and alternative livelihood opportunities it could very well be beneficial to Papuan livelihoods, but ensuring that the potential benefits of these policies are equally distributed and do not reinforce existing power imbalances will require additional efforts on the part of the national government.

Implement all recommendations listed under the passive use scenario.

Any form of active promotion of MBLs must be paired with significant investments into the preservation of TEK and other customary practices.

Promote the establishment of community-led tourism offices, where communities outline the objectives of tourism projects and retain control over all aspects of ecotourism, including accommodation, transport, tours, and catering.

Promote integration of tourism activities regarding wildlife watching and indigenous customs to maximize the educational potential of ecotourism.

Invest in capacity building programs for the local population, including foreign language learning and hospitality training.

Raise awareness among communities of the potential cultural, environmental, and identity impacts that ecotourism may have on local peoples.

Establish a national platform of village- and homestays where hosts can engage in mutual learning (both online and in training events) and tourists can easily find and contact hosts.

Require communities to reinvest part of the obtained benefits from MBL schemes into forest conservation, reforestation, and/or cultural heritage events.

Impose a carbon tax for any foreign tourist entering the country by means of air travel, which may be used to fund reforestation and/or conservation projects throughout the country or contribute to the government's support for community forestry projects.

Create a Traditional Knowledge Digital Library (TKDL) database.

Implement a tender system for bioprospecting arrangements that identifies potential genetic resources or practices of willing communities and links them to bioprospecting businesses.

8. Logging

West Papuan communities have always cut down trees to meet various basic needs, using timber for heavy construction, tool making, and as a fuel source. Similarly, multinational industries rely on the extraction of timber from forests as a raw material for heavy construction and the furniture and paper industries. The commercial extraction and economic valorization of timber can be a good way to generate social benefits and tax revenue for Papuans. However, any extraction of forest resources that exceeds the frequency of natural processes also poses a potential threat to the long-term health of forest environments and the survival of their inhabitants. Logging and forest management interventions affect the composition of forests and the habitats they help form quite directly, as live trees and dead woody debris (necromass) each fulfil a variety of important roles in their environments. In particular, the stability and composition of soils and the retention of canopies affect the maintenance of ecosystem services^{28,176–179}, whereas the regenerative ability of forests depends on the availability of seed-trees nearby^{28,179}, as well as on the composition of the soil^{28,180}.

Many different methods of timber production and extraction have been generated over the course of millennia, including different felling regimes, management interventions, and yarding methods, all of which have their unique set of economic, environmental, and ecological implications (see Annex 4). Implications of logging methods are also dependent on the geography and ecology of a particular location, which warrants varied approaches that are only fundamentally understood by communities who have customarily inhabited these forests. Diversity of forest management approaches is thus fundamental to West Papua's cultural and ecological richness and should be encouraged. To account for the various tradeoffs between economic opportunity and the security of biodiversity and ecological processes and to explore the different options available to the West Papuan government to prioritize economic opportunities over conservation goals we outline three different scenarios pertaining to logging regulations: 1) no industrial activity, 2) community industry, 3) restricted industry. Based on the natural gap dynamics of tropical forests and our multi-factor analysis of the impacts of different logging methods, we also recommend a maximum patch size and that certain forest management techniques are universally avoided to prevent severe and long-lasting damage to forests.

The state should prohibit the clearing of any patch of forest larger than 1000 m² (0.1 ha) and of patches not surrounded by intact primary forests.

The state should prohibit the use of bulldozers and high lead yarders in logging operations.

The state should prohibit mechanical scarification of soils in forest environments.

Exceptions may apply to reforestation projects of agricultural lands and roads where forest was cleared before the establishment of the West Papuan Green State.

The state should prohibit the application of chemical herbicides and pesticides in all logging operations, tree plantations, and agroforestry projects.

The state should prohibit the construction of logging roads and skid trails, as well as the use of helicopters, farm tractors, and skidders and any other form of motorized heavy machinery in forests, unless their use is explicitly planned for in the forest management plan and has received official approval by the community managing the territory as well as by the national forest authority.

The state should prohibit the practices of enrichment planting, thinning, controlled burning, vine cutting, and tree girdling by CCPs, unless their use is explicitly planned for according to RIL principles in the forest management plan, and has received official approval of the community managing the territory as well as by the national forest authority.

The state should allow for the use of enrichment planting, thinning, controlled burning, vine cutting, and tree girdling by CFEs and non-incorporated communities if they are part of traditional practices and applied selectively.

The restrictions listed here must not be enforced when logging activities solely pertain to subsistence use, nor in any community that does not engage in commercial activities.

8.1 Logging scenarios

Scenario 1: no industrial activity

Strengths - Limits risk of deforestation & forest degradation - Low-cost monitoring and enforcement	Weaknesses - Limits income generation and livelihood diversification - Slightly limits self-determination of communities
Opportunities - Retention of culture & identity, including TEK - Forest protection through customary governance	Threats - Illicit extraction of timber becomes more likely - Conversion for agriculture becomes more likely

The safest option in terms of protecting the biodiversity and ecological services that forests provide would be to ban all commercial exploitation of timber. In this scenario, communities would still be allowed to cut down trees to meet their immediate needs including fuel and construction, but the trade and export of timber are banned in order to prevent deforestation as a result of global market pressures. While this approach is closest to the customary use of forests that has allowed the forests of West Papua to sustain themselves in the state they are in now, embracing this approach cannot guarantee that deforestation is halted in the current global context. Eliminating all legal pathways to enjoy economic opportunities from timber risks driving loggers to black markets instead^{141,181}, and could be an incentive to convert forests to agricultural land for the cultivation of cash-crops, especially if few other economic opportunities exist.

Prohibit all logging for commercial activity for corporations, communities, and individuals.

Prohibit the internal trade of uncut logs, as well as the export of all wood products, high-value artisanal products and art and wood used for research purposes may be exempt.

Adopt *community industry* or *no industrial activity* scenario for land use conversion.

Scenario 2: community industry

Strengths - Greatest self-determination for communities - Limits foreign market & corporate influence	Weaknesses Organizational & administrative burden on communities
Opportunities - Income generation for state and households - Livelihood diversification for households	Threats - Risk of dependence on third party traders - Increased inequality between & within communities

To allow for communities to exploit the timber within their territories for economic opportunities without exposing West Papuan forests to excessive market pressures, the government may choose to ban or restrict foreign companies from being involved in the timber trade and for community-based commercial logging to be limited to small-scale operations that mimic natural disturbances of the forest. Limiting the

influence of foreign companies in the timber industry could result in the best balance between allowing economic benefits to flow to communities and ensuring the survival of more-than-human beings but instates a large administrative hurdle for communities to tackle. A lack of market knowledge and legal and technical capacity amongst community members puts communities at risk of becoming overly dependent on third-party traders and of magnifying power differences between communities and community-members.

Prohibit corporations and CCPs from engaging in logging operations.

Allow CFEs, communities without a business registration, and individuals to engage in small-scale selective logging operations.

Allow for the sale and trade of logs and wood products.

Prohibit foreign corporations from investing in logging and land governance projects.

Adopt *community industry* or *no industrial activity* scenario for land use conversion.

Scenario 3: restricted industry

Strengths
• Greatest income generation for state

Weaknesses
• Costly administrative, monitoring, and enforcement burden on state

Opportunities
• Livelihood diversification for households • Capital investments in community services • Easy access to foreign timber markets

Threats
• Degradation of culture & identity, including TEK • Decline of NTFP availability & food security • Degradation of forest health & biodiversity • Increased risk of corruption & elite capture • Increased inequality between & within communities

Allowing logging by all actors will maximize the economic opportunities for West Papua as a country but will require stringent conditions in terms of benefit sharing and logging methods in order to ensure that the needs of human communities and more-than-human forest species are met. Even under strict conditions, allowing logging by companies as well as communities increases the likelihood that species will go extinct or become more threatened, that NTFP availability for humans is reduced, that livelihoods become more market-oriented and TEK is lost, and that conflicts between communities and within communities or CCPs arise. The increased investments and foreign financial interest in West Papuan forests also increase the risk of corruption and use of heavy machinery in logging operations.

Allow CCPs, CFEs, communities without a business registration, and individuals to engage in logging operations.

Allow for the sale and trade of logs and wood products.

9. Competing land uses

The main threat to forests is not the direct extraction of forest resources, but the conversion of forests to alternative land uses, also known as land use and cover change (LUCC). Conversion of land allows people to grow certain crops more effectively, extract valuable minerals from the soil, or build infrastructure that facilitates trade, which can all create new income streams for the people exploiting it. However, when forests are converted to land for alternative uses they reduce or eliminate the habitats and materials that forests provide, which can also have detrimental consequences for the livelihoods of local people and for the species that are able to live in this environment. LUCC regularly lead to conflicts between parties with different interests, especially when overlapping land rights, corruption, economic inequality, migration, high dependence on agriculture, and neglected Indigenous land rights are involved¹⁰⁹. Policies regarding the use of land for agriculture, mining, infrastructure construction, and urban expansion thus affect the state of forests in important ways⁷¹ (see Annex 5 for an elaborate comparative analysis of these impacts).

To achieve a diversified forest landscape that balances local priorities appropriately and avoids land conflicts in a modern state and globalized world requires strong local participation, including secure land tenure and inclusive decision-making processes, as well as a prioritization of shifting and small-scale developments and land conversion over permanent, large-scale, and monocultured land uses. However, the national government also has a responsibility to consider the effect of national policies on local forest uses, and it is necessary to weigh and balance different priorities such as generating tax revenue and ensuring food security. Since the choices made here must be informed by current political preferences, we propose three different scenarios that might be employed to balance the different needs of West Papua as it changes over time: 1) no industrial activity, 2) community industry, 3) restricted industry.

The state should prohibit large-scale (patches over 1000 m² (0.1 ha) or exceeding 4% of the territory's surface area) monocultural plantations and open-field agriculture.

The state should allow agroforestry practices that retain the existing canopy and grow multiple native crops in conjunction, but such agriculture must not cover more than 30% of a territory.

The state should only allow open field agriculture (including swidden) for commercial purposes and/or using cash-crops if they are surrounded by natural forest and returned to a fallow state within 5 years of establishment and fallow for longer than 10 years afterwards.

The state should only allow monocultured tree plantations and perennial food crops if they are community-owned, on small patches surrounded by natural forest, and returned to a forest state after cultivation.

The state should only allow cash-crop agroforestry if they are not preceded by clearing of land and canopy cover is maintained throughout transition, and if multiple tree species that are native and/or multifunctional are cultivated.

The state should promote practices that retain the existing canopy, practice intercropping, or use native plants.

9.1 Land use conversion scenarios

Scenario 1: no industrial activity

Strengths • Greatest maintenance of forest cover • Based on customary practices • Limits risk of land grabbing & corporate abuse	Weaknesses • Limits income generation & livelihood diversification • Costly monitoring & enforcement burden on state • Slightly limits self-determination of communities
Opportunities • Retention of culture & identity, including TEK • Forest protection through customary governance	Threats • (Illicit) Extraction & trade of forest resources becomes more likely

Banning forest conversion for commercial purposes would maximize the maintenance of forest cover and consequently the availability of customary livelihoods and the likelihood of survival of forest-dependent species. In this scenario, the use of home gardens, swidden fields, and agroforests to grow crops for consumption and local trade would remain the main source of livelihood provision, which would protect the TEK and customary ways of life of Papuan peoples and shield Papuan food security from the fluctuations and pressures of global markets, but also limit the options for livelihood diversification and economic growth. There is little risk involved with this approach, but restricting industrial activities in this way would also allow for little to no tax revenue to be generated. Additionally, it would require significant resources to enforce because, unlike restricting logging which can be enforced through trade controls, land conversion happens in-situ and enforcement can only be done if all forest land is monitored on the ground in a detailed and meaningful way.

Prohibit all forest land conversions for commercial activity for corporations, communities, and individuals.

Allow trade of cash- and food crops cultivated in forest environments on local markets only.

Complement this legislation by researching and investing in a(n inter)national traceability scheme for agricultural crops grown in existing agricultural and other non-forested areas.

Scenario 2: community industry

Strengths • Greatest self-determination for communities • Based on customary practices • Limits foreign market & corporate influence • Low-cost monitoring and enforcement	Weaknesses • Limits income generation for state
Opportunities • Income generation & livelihood diversification for households • Organic dissemination and selection of beneficial arrangements	Threats • Increased inequality between communities • Risk of dependence on third party traders • Slight risk of decline of NTFP availability & forest health, and homogenization of landscapes

Prohibiting land conversion at a large scale and by corporate actors would allow communities to engage in small-scale cash-cropping and thus create new economic opportunities for Papuans. This approach is in line with common traditional livelihoods¹⁸², but it might indirectly expose communities to increased

market pressures that could increase market-oriented thinking and create harmful dependencies on intermediaries. While small-scale and shifting agriculture practiced in the past have been able to maintain a high level of biodiversity in West Papua, engagement with global markets might still have an effect on the composition of forests and on the species that rely on these environments. An approach restricting corporate and large-scale forest conversion is likely to maximize the degree of self-determination for communities, and would require little government capacity to enforce, but have limited potential for the generation of tax revenue.

Prohibit corporations and CCPs from engaging in agricultural and agroforestry projects on any land that is forested as of the date of independence.

Allow CFEs, communities without a business registration, and individuals to engage in small-scale land conversions.

Allow small-scale (<0.1 ha) land conversions for cash-crop agroforestry, cash-crop plantations, and perennial food crops (i.e. orchards) using native and naturalized plants.

Allow medium-scale (<1 ha) agroforestry projects only if they use native or naturalized plants and retain the existing canopy (i.e. do not cut trees over 20m in height and retain a total canopy cover of at least 70%)

Allow trade of cash- and food crops on local and international markets.

Scenario 3: restricted industry

Strengths	Weaknesses
Greatest income generation for state	Costly administrative & compensatory burden on state
Opportunities	Threats
- Livelihood diversification for households - Capital investments in community services - Easy access to foreign cash-crop markets	- Degradation of culture & identity, including TEK - Decline of NTFP availability & food security - Degradation of forest health & biodiversity - Increased risk of corruption & elite capture - Increased inequality between & within communities

Allowing for land conversion by all actors who meet a set of certain conditions would maximize the potential income of land generation, but also increase the risk for losing forest-dependent species and ecosystem services that form an important part of the customary livelihoods and environmental health of West Papua. Allowing corporations and CCPs to engage in forest conversion will likely have a homogenizing effect on the forest landscape and thus pose a risk to biodiversity as well as to communities' self-determination. A universal policy approach to forest land-conversion will also allow for a degree of industrialization of forests and magnify market-based thinking in West Papuan society, so the maintenance of cultural traditions and sustainable forest governance in this scenario will require more active involvement of the government. Industrializing the forest in this way, however, will also generate more government funds from tax revenue that may be used to invest in social protection and cultural enrichment projects.

Allow CCPs, CFEs, communities without a business registration, and individuals to engage in small-scale land conversions.

Allow small-scale (<0.1 ha) land conversions for cash-crop agroforestry, cash-crop plantations, and perennial food crops (i.e. orchards) using native and naturalized plants.

Allow medium-scale (<1 ha) agroforestry projects including those that use exotic plants and cash-crops only if they retain the existing canopy (i.e. do not cut trees over 20m in height and retain a total canopy cover of at least 70%.

Allow trade of cash- and food crops on local and international markets.

9.2 *Competing policies and government projects*

Aside from policies directly pertaining to forest conversion and the commercial use of currently forest land, the government must also avoid certain policies in the fields of agriculture, infrastructure, and mining that are not directly linked to forest conversion but have indirect effects on forests, as well as consider the effects of state-led infrastructure projects on forests and forest communities.

The state must have the duty to safeguard traditional agricultural methods including swidden farming and home gardens.

The state must avoid policies and discourage practices promoting agricultural intensification of any sort, including the shortening of fallow times.

If the state grants subsidies for agriculture, it must consider applying these to maintained primary and secondary forests as well.

The state should avoid the subsidization of specific cash-crops and prioritize native species that grow well in community with other plants and a.) provide multiple benefits for humans such as fruit and timber (e.g. Galip, Okari, Breadfruit, Massoia), or b.) give a high return to land (e.g. Sandal- or Agarwood; hardwood like Merbau, Matoa, Kwila, Hoop or Klinkii pines).

The state should develop a new assessment tool for food security and poverty that includes at least NTFPs, nutritional quality, and health status.

The state should discourage population growth and resettlement in regions with high forest cover and low population density.

The state must restrict the construction of (logging) roads and settlements in intact forests.

The state must prioritize upgrading existing road capacity over building new roads and widening roads.

The state should employ strategic road development planning that integrates a measure of relative forest value including human and more-than-human values.

The state must collaborate with and find approval from the communities affected by infrastructure projects during the planning process.

The state should disincentivize private vehicle ownership by investing in public transport networks and in pedestrian and cycling infrastructure.

The state should promote outfitting urban areas with green spaces and permeable pavements.

The state should prohibit industrial mining projects.

The state should prohibit the creation of new pastureland.

10. Threatened species

Many of the species that are unique to West Papua or that have an important role in meeting the nutritional, medicinal, and cultural needs of Papuan humans are under threat of extinction^{25,27,183,184}. Many of these species only occur under unique conditions and have very limited ranges due to the isolated location of New Guinea and the great topographical variance on the island^{14,61,185}. The highest rates of species richness are found in highland areas^{12,186}, but the species most at risk of extinction often occupy larger patches of lowland forest where habitats are declining at a more rapid pace³⁴. The most important threats to the survival of more-than-human species in West Papua are habitat loss and hunting or gathering pressure²⁷. Globally, the most efficient and cost-effective conservation method to protect threatened species is the establishment of Indigenous land rights in Melanesia¹⁸⁷, but supplementary policies may still improve the effectiveness of species protection by Papuan communities.

10.1 Extraction

Despite an elaborate history of hunting in Papua²³, hunting nowadays forms a real threat to the conservation of some forest species of fauna. Although the threats to flora in West Papua are less well-studied, data from PNG¹⁸⁸ and select plant species⁸⁴ suggests that the same can be said for foraging plant resources. Due to the importance of many of these species for human livelihoods in Papua, banning the use of all at-risk species would be unjust and unlikely to result in effective conservation outcomes, so finding sustainable methods to harvest and govern NTFPs is essential for sustaining the biological and cultural richness of West Papua. TEK generally forms a good basis for shaping harvesting and extraction practices that are able to sustain populations over the long-term¹⁸, yet certain biases and dependencies can also create blind spots for this approach^{21,84}, especially if new environmental pressures such as pollution, increased accessibility due to roads, population growth, and new hunting techniques increase the rate of change^{26,183,189,190}. Migrant trappers and hunters from outside the community have increased harvesting pressures as well^{26,183,189}. A sustainable approach to protecting forest species from extinction due to overextraction is thus one that combines customary land rights, traditional harvesting techniques, and customary hunting restrictions with scientific data and monitoring technologies. As communities and scientists both have a vested interest in the conservation of species and important gaps in their knowledge on how to do so, collaborations should be fruitful for all parties involved.

The state should embrace and promote customary practices of forest protection and resource governance such as *sasi* and *teter* to protect vulnerable species locally.

The state should offer, upon request, technical assistance to communities for aid with monitoring and determining sustainable offtake levels of nutritionally or culturally important but vulnerable species.

The state should encourage knowledge sharing and capacity building between communities, NGOs, and scientists to develop a more complete picture of landscape-level conservation needs, increase the scientific data available on species occurrence and behavior, and to find locally appropriate management interventions when necessary.

The state should prohibit the use of snare traps and shotguns in hunting.

10.2 Protected areas

Protected areas (PAs) are a mainstay in the conservation of species, but they have often been used for the exclusion of Indigenous and local human populations from the resources they rely on for their livelihoods. Although it can be argued that certain forests and species have ‘high-conservation-value’, prioritizing certain areas over others in terms of the environmental protection they are to receive is a complex and heavily political undertaking, especially considering the importance of connectivity of habitats (as opposed to fragmented well-preserved areas)^{34,191}, the fact that forest communities cannot rely solely on species and habitats that are conserved elsewhere, and that technical reports identified

conflicting priorities^{185,192,193}. For the protection of species across the full range of forest environments in West Papua, all ecotypes and elevational gradients should be represented within PAs³⁴, and to be effective, conservation efforts need to be paired with participatory mapping and livelihood improvement schemes and be implemented at a community level using customary rules and practices^{21,34,70}. Overall, intact forest landscapes that have a great degree of connectivity and include different successional stages should cover at least 70% of the total territory and cover all different ecotopes and elevational zones that exist within the area^{34,131,194}, which is likely not feasible as an active policy for the West Papuan government and might not be beneficial to Papuan people. Instead, an integrated approach designating all forested customary territories as **Indigenous and Community Conserved Areas (ICCAs)** is most likely to balance the protection of beings in West Papua.

The state should assign all currently forested land as Indigenous and Community Conserved Areas (ICCAs).

In the long-term and after consultation of the affected communities, the state should pursue the conversion of all existing protected areas to ICCAs.

The state should recognize secondary forest stands and fallow stands as forests in early stages of succession rather than as degraded lands.

10.3 CITES

The trade in threatened species for the pet-trade or traditional medicine can pose a serious threat to the maintenance of healthy wild populations of certain forest species, which is increasingly true in West Papua^{183,190,195}. CITES is the international convention that regulates the trade of these species. Becoming a signatory to the CITES convention requires implementing national legislation to enforce its requirements. Since CITES has been relatively effective at halting the commercial exploitation of threatened species across the world as well as within West Papua^{84,141,181}, it would benefit West Papua to implement such laws.

West Papua should become party to the CITES treaty and implement the required legislation on a national level.

The state should prohibit the trapping of and trade in any live wildlife or animal body parts for international markets. Exceptions may include extraction for research.

11. Carbon stocks & watersheds

The carbon stocks contained within forests are of particular importance for meeting the GSV's objectives in terms of mitigating global climate change. Forests on peat-soils and mangrove forests have an important role to play in relation to climate mitigation and adaptation^{193,196–199}. Peat soils contain by far the highest amount of carbon of any ecosystem due to its waterlogged conditions, and as long as a high level of moisture is maintained atmospheric carbon can be stored permanently. The biggest drivers that increase carbon release from peat, therefore, are the drainage and burning of peat soils^{196,198,200}. Much like the global climate, the governance of watersheds poses an additional challenge because they are generally transboundary concerns. Mangroves and riparian forests are very important for flood control and the maintenance of water quality by preventing erosion and increasing sedimentation²⁰¹. Since watersheds present large connecting systems that flow through different territories, the governance decisions of one community regarding their riparian forests may fundamentally affect other communities downstream. Although communities play an important role in, and are often capable of, managing peat soils and watersheds sustainably, coordination and collaboration can prevent governance gaps^{108,202,203}. As such, as the principle of subsidiarity suggests, there needs to be governance regarding carbon stocks and watersheds that exceeds the community scale.

The state should impose the responsibility of each person, community, and government body to protect the carbon stocks and water bodies within a territory.

The state should prohibit industrial and commercial development of plantations on peat soils.

The state should prohibit the drainage of peat soils and major earthworks in and around forests on peat soils.

The state should invest in raising awareness and finding alternative practices for peat conversion and swidden agriculture for communities living on peat soils and/or in mangroves.

The state should allow and may promote the growing and trade of Sago on peat soils under the condition that the area stays sufficiently inundated.

The state should invest in research and development of other forms of paludiculture practices by Papuan communities.

11.1 Protected areas

Since the interest of conserving mangrove and riparian forests and forests on peat soils is far greater than the interests of the local communities alone, the national government may wish to implement policies that grant additional protection to these areas. One common solution is instating riparian and coastal buffer zones of at least 50 meters where trees cannot be cut^{201,204,205}. Certain activities may also be banned on peat soils, for example by restricting conversion of forests on peat soils to only be allowed for paludiculture, a form of agriculture where peat soils are not drained, which is suitable for the cultivation of Sago^{206,207}. However, such protection should not impede the self-determination of local communities and compensation for limiting the activities in these areas may be warranted.

The state should prohibit logging and land conversion for commercial purposes in forests within coastal and riverine buffer zones (100m from the waterside at high tide, including mangrove forests).

If hard restrictions are placed on cultivation on peat soils, High Carbon Stock areas, and/or mangrove forests, the state should compensate communities on peat soils for missed potential opportunities from commercial exploitation, possibly through a fund financed by carbon taxes from international travel.

Concluding note

This policy brief has tried to provide recommendations for suitable forest policy options for West Papua from the perspective of, and as informed by, the academic fields of forestry, conservation, and legal studies. However, the very premise of this work is in some ways at odds with its main conclusions: that engaged communities with strong institutions and specialized socio-environmental knowledge, rather than distant scientific researchers or policy makers, should be deciding on which policies apply to their territories.

In addition, it will be the political economy of West Papua, rather than its individual policies, that determines what happens inside the nations' forests. The main purpose of West Papuan forest policy must thus be to prevent Papuans' well-being becoming dependent on extractive industries which degrade the forest. Consequently, the state must also avoid depending on tax income from forests to fund its social programs.

As such, we would like to reiterate that most recommendations in this report should be taken as well-studied and comprehensive but somewhat arbitrary policy approaches that revolve around four core tenets:

- 1) communities must be granted secure land tenure and full autonomy over decision-making regarding the use of land and forests within their respective territory,
- 2) the state must establish inclusive decision-making for all by guaranteeing fair representation of marginalized groups in regional and national governments, as well as by promoting the importance of women and other marginalized sub-groups in decision-making at the community level,
- 3) the state must limit the influence of corporations and foreign markets over communities' autonomous decision-making by imposing social and environmental requirements to their operation and controlling international trade,
- 4) the state must limit the effect of other policy fields on forests and its associated livelihoods, in particular by imposing strong restrictions on road construction, industrial agriculture, and mining.

No nation-state on earth is currently able to fulfill all the objectives that West Papua's Green State Vision has outlined, so achieving these objectives necessarily requires the co-creation of new approaches through in-depth and meaningful engagement and consensus-building between all actors and level of government in Papuan society. We hope that this policy brief can contribute to establishing a framework for such novel decision-making processes. If West Papua does succeed in achieving its vision, it may truly provide guidance and leadership to other states in addressing the climatic, ecological, and democratic crises faced across the world.

Compiled list of recommendations

Legal framework

Rights and responsibilities

Justice and governance should be based on responsibilities and duties of groups of individuals within an interdependent community of living beings, rather than on individual rights.

‘Rights’ in national policy should only be granted to collective entities and not be vested in individual humans.

Rights-based approaches to national and community governance should be understood as a strategic tool rather than an inherent property of existence and may only be embraced if the rights in question are held collectively.

Justice

Legal procedures and punishments should focus on the healing of human and more-than-human victims, which requires facilitating dialogue and storytelling between victims and perpetrators, and in the context of forests may mean compensation for harms to human health and lost livelihoods as well as ecosystem restoration projects.

The state should pursue, and implement mechanisms to achieve, relational and restorative forms of justice, rather than retributive justice, in response to (environmental) violations.

Governance structure

The state should employ a system of polycentric governance, where governance duties are executed and services are provided by multiple governance centers, and where the national government only provides an overarching system of rules and goals.

The state should impose different restrictions on harvest for subsistence and community use, and extraction for commercial use and exports.

The state must recognize its own duty to implement and enforce customary law in recognized customary territories.

The state must establish the legal authority of customary courts in national law, and nest customary courts at the community and tribe level inside the national judicial system.

Division of responsibilities and duties

Governance duties and arrangements related to forest lands should be divided based on the principle of subsidiarity.

Timber and NTFP harvesting activities and interventions relating to natural resource and land management in general should be governed by local communities.

The state must not impose legally binding restrictions or responsibilities on communities’ subsistence use of forest resources.

The state must not enforce regulations within customary territories unless support is specifically requested by the community.

The state must not undertake any large-scale land zoning or planning on forest lands.

The state should have the responsibility to ensure inclusive decision-making processes at all levels of government.

The state may promote certain practices and organizational aspects relating to forest governance, such as the deliberative and transparent strengthening of local institutions and participation in national or intertribal associations of forest producers.

The state should prioritize customary methods of conflict resolution but provide third-party mediation and multi-stakeholder dialogues for when these are demonstrably exhausted.

The state should allow a national supreme court to hear cases that are appealed in customary and tribal courts and interpret the application of customary laws in doing so.

The state should support inter-tribal collaboration and organization with regard to watershed protection and species conservation by contributing financial and technical resources, as well as hosting inter-tribal events aimed at knowledge-sharing.

The state should create a national institute for forest health to monitor the status of threatened forest species and habitat connectivity at a landscape scale. This institute may relay observed and scientific information to affected communities and provide capacity building programs to support community management plans but must not have a legal mandate to impose specific restrictions or prohibitions.

Any national or regional advisory board that oversees landscape-level issues should be made up of representatives from different tribes, may operate as a working group of the customary council, and should be financially supported by NGOs, members, and/or taxes, but should operate independently from the national government.

Funding government programs and services

The state may levee moderate sales taxes on commercial activities that rely on the use of or extraction from forest environments and trade tariffs on exports.

The state may generate state income through bilateral agreements with other states only if they do not shape national legislation nor limit local decision-making or subsistence use of forests.

The state should prioritize non-forest sources of state income generation over forest-dependent sources of funding.

Land Tenure

Rights to what?

The state should have the duty to guarantee the ability to govern ancestral lands to Indigenous communities by recognizing a bundle of Indigenous land rights in the most fundamental levels of West Papuan law, i.e. within the constitution, and for an indefinite duration of time.

The state should grant the ownership and usufruct rights of land and all resources contained within a territory to their respective customary communities, including minerals in the soil.

The bundle of land rights for Indigenous communities should include the rights to 1) land access; 2) resource extraction; 3) environmental management; 4) exclusion of outsiders; 5) alienation of aforementioned rights by sale (to the state or other communities) or temporary lease; 6) due process and compensation in case of the revocation of aforementioned rights.

The state should prohibit foreign corporations from owning forest land in West Papua. As such, the right to alienation of customary land should be restricted to sale within or between customary communities, and management and extraction rights may be leased out to corporations for a maximum of 50 years.

National law should apply a separability principle between the land and the buildings and plants on it: individuals and corporations may borrow or lease the rights to enjoy what the land produces from the land-owning community but may not own the land itself.

Contested areas and areas with overlapping land claims may not be exploited for commercial purposes and external parties must be prohibited from entering into agreements or contracts (including sale or lease) that apply to these areas.

Since land rights are owned by and assigned to the collective community and not individuals, any benefits derived from the exploitation of forests must be subject to equal benefit sharing.

Recognition process

All land in West Papua should be recognized as belonging to customary communities and no forest land should be owned by the state unless it is purchased from the relevant community.

Recognition of ancestral lands should be established in law as the default position and not be dependent on formalization.

The validity of land rights should precede official documentation and formalization of the specific customary territories.

The state should avoid overly prescriptive paperwork and administrative requirements for the recognition procedures for customary territories.

The state should pursue the use of dynamic and multilayered mapping techniques.

The state should support the participatory mapping of customary territories with financial and technological capacity.

Communities should own and have free access to maps of their territory, and they should have an active role in choosing which layers are included whenever these maps are applied.

The state should promote the involvement of outsiders (i.e. non-Indigenous humans present in West Papua) in customary practices and rituals, as well as communal protection

of the environment, and interethnic and intergenerational dialogue as tools for identity building, peacebuilding, and conflict resolution.

Inclusive decision-making & social protection

Self-determination and ontological pluralism

The state must be cognizant of the divergent realities that exist within West Papua and avoid naturalizing its own reality and legal definitions by using legal language and engaging with communities in ways that allow for different interpretations.

The state must have the duty to engage with a broad cross-section of different social groups in a community when it collaborates with or seeks the free, prior, and informed consent of a community and may do so by consulting their respective informal leaders or by randomly selecting members of each group. This includes respecting the right of a community to withhold consent for any development that may affect their collective rights.

The state should allow the establishment of CFEs for the commercial exploitation of forest resources and the representation of the community in national and international markets.

The state should support communities in democratizing existing and forming novel management institutions through organizational capacity building and mediation.

Benefits sharing and money in politics

Equal benefit sharing should constitute either direct payments to all adult community members individuals or collection in a community fund that is collectively managed through participatory budgeting or a democratically elected management board.

Payouts of benefits may not be appropriated or distributed by the national government.

The state must prohibit private payments to military, police, and security forces and require transparent budgeting for the military and the police force.

The state must restrict private donations to political campaigns and require transparent reporting on campaign finances.

The state must ban the practice of vote buying and support this prohibition with public awareness-raising campaigns and by monitoring candidates and corporate conduct.

Women in forest policy

The state must guarantee the fair representation of women and tribal minorities in national and regional governments.

The state must establish equality between women and men with regards to the ability to hold and own land in the most fundamental levels of West Papuan law, notably the constitution.

The state should actively promote the inclusion of women in deliberative consensus-seeking discussions (e.g. *para-para*) within communities.

Women's participation in decision-making processes at the community level may take the form of women-only consultations or forums, alongside consultations that involve both men and women together.

CFEs and landowner companies must be overseen by a broad board of community members in which women and men are represented equally and hold comparably influential positions.

Traditional Ecological Knowledge

The state should encourage inter-community organization in forest-producer associations and networks.

The state must legislate on the customary communities' rights to intellectual property, including traditional knowledge with regards to the use and processing of forest resources.

The state must avoid human-nature dualism in government discourse, policy, and law.

The state must avoid standardization of education and include TEK and local languages in the public educational curriculum.

The state should invest in capacity building for indigenous artisanal practices and skills.

Corporate-Community Partnerships

Procedure of establishment

Communities may enter into a partnership with an external (foreign or domestic) company (CCP), either as a CFE or as a community without a registered business license.

Consent-seeking procedures and Social Responsibility Agreements should be bottom-up rather than top-down: a community needs to consent before any licensing may be granted.

The state may employ a community-led or state-led application procedure for the establishment of CCPs.

Agreements within a CCP should be made individually with each community involved.

CCPs may not exploit resources located on land that is disputed by neighboring tribes or clans.

Contents of CCP agreements

The state must be responsible for safeguarding the just treatment and consent of communities in CCPs and may require the involved company to provide documentation to this end.

All documentation and reporting that the state requires must be submittable in multiple forms (e.g. written, video, audio, etc.) to ensure their accessibility for non-literate people.

Companies applying for a CCP must have the duty to provide proof that broad awareness and support of the project plans exists among both male and female community members.

CCP agreements should include: 1) a community-made map of the territory; 2) a statement of social obligations specifying the culturally specific sites, practices, and dates to be respected; 3) a benefit-sharing and redistribution plan; 4) a forest management plan; 5) a description of the division of responsibilities; 5) a plan for the continued engagement between the partners in a way that is appropriate to the community's customs and capabilities; 6) a consideration of the long-term future (at least 50 years); 7) a clear description of mechanisms for conflict resolution; 8) an explicit agreement on lease length and terms of termination.

CCP forest management plans for extractive activities should include: 1) a protected area that covers a significant portion of the primary forest, connects to forests on neighboring territories, and represents all ecosystems within the territory; 2) a harvesting plan according to RIL guidelines; 3) a full stand inventory; 4) a management intervention plan that includes a strategy to minimize negative impacts on NTFPs including an indication of expected social and ecological benefits; 5) a commitment to the registration of all extracted resources in forest product registries.

Reporting, monitoring, and enforcement

Companies in a CCP should have the duty to report biannually on activities relating to company-community relations and on stand stocks of timber and NTFPs.

The state should create the position of an environmental ombudsperson that operates independently from the government and is responsible for ensuring national and local policy is in line with securing inclusive and community-level forest and natural resource governance.

Forest product registries

The state should create a national forest products registry system for monitoring the flows of all timber, animal, and plant products extracted from forests intended for commercial trade.

Communities that wish to sell their forest products to an international market should be required to maintain and report on a register for each class of export product.

Sawmills should be required to maintain and report a register that notes the origins of the timber they receive and to what products each of these converted.

Traders should be required to be able to present a full registry of the origins for each product they are exporting.

Border & customs agents should have the duty to copy and enter the relevant export data into a national registry.

The state should allocate the responsibility of cross-referencing the reported registries to an internal department.

The state should establish and delegate funds to a separate investigative department, or hire a third-party, to conduct occasional audits at all steps within the forest resource export chains.

Trade and export

The international borders of West Papua, on land and at sea, must be a central aspect of monitoring and enforcement of forest use and resource extraction by the national government.

The state must make significant investments in setting up a strong and transparent border customs system.

The state should pursue full transparency and traceability of forest product supply chains to facilitate ethical trade, and to limit corruption and environmental abuses.

Export licenses for forest products should only be granted to parties that can present evidence of where the transported products were sourced at a community-level.

The state should impose fines on illegal exports that are proportional to the captured cargo's value, for which the minimum fee exceeds at least the total value of the confiscated products.

Confiscated goods and charged fines should be returned to the community from whom the goods were stolen if possible or used by a government agency to support communities active in forest product trade.

Companies that violate trade regulations and their owners should be denied access to future export licenses and efforts should be made to prosecute the executives higher up the chain.

The state should also collaborate with the PNG government to coordinate legislation and oversee the land border to avoid leakage of forest products across the border.

The state should ban the export of uncut logs.

The state may impose an export tax on all forest products transported across the border.

Market-based Instruments

The state must simplify legislation, regulatory language, administrative procedures, and communication with communities regarding MBIs as much as possible.

The state should facilitate the engagement of communities with any MBIs by means of legal capacity building and offer third-party mediation where necessary.

The state should invest in social learning for its employees to achieve a clearer understanding and better definitions of local customary and legal institutions, knowledges, and needs.

ABS & Nagoya Protocol

The state should become party to the Nagoya protocol and implement the required legislation on a national level.

The state must be responsible for protecting the ability of Papuans to use and trade all customary crops and genetic resources from encroachment by foreign patents.

The state must require proper reporting on PIC and MAT when granting patents on natural and genetic resources, and promote this practice internationally.

Scenario 1: Restriction

Ban international PES schemes, including REDD+.

Allow communities to engage in community-based ecotourism projects and bioprospecting.

Prohibit foreign and corporate investments in ecotourism business and infrastructure.

Scenario 2: passive use

Allow communities to engage in ecotourism, PES schemes (incl. REDD+), and bioprospecting.

Promote the use of inclusive and pro-poor policies in PES schemes.

Include traditional land uses (e.g. swidden fallows and agroforests with maintained canopies) in REDD+ and other PES schemes.

Require the establishment of a CCP for all PES schemes and for external parties partaking in ecotourism, including hotel owners, tour operators, and travel agents.

Pursue the formation of a genetic resources ‘cartel’ with neighboring countries to protect genetic resources and customary practices from a ‘race to the bottom’.

Scenario 3: active promotion

Implement all recommendations listed under the passive use scenario.

Any form of active promotion of MBIs must be paired with significant investments into the preservation of TEK and other customary practices.

Promote the establishment of community-led tourism offices, where communities outline the objectives of tourism projects and retain control over all aspects of ecotourism, including accommodation, transport, tours, and catering.

Promote integration of tourism activities regarding wildlife watching and Indigenous customs to maximize the educational potential of ecotourism.

Invest in capacity building programs for the local population, including foreign language learning and hospitality training.

Raise awareness among communities of the potential cultural, environmental, and identity impacts that ecotourism may have on local peoples.

Establish a national platform of village- and homestays where hosts can engage in mutual learning (both online and in training events) and tourists can easily find and contact hosts.

Require communities to reinvest part of the obtained benefits from MBI schemes into forest conservation, reforestation, and/or cultural heritage events.

Impose a carbon tax for any foreign tourist entering the country by means of air travel, which may be used to fund reforestation and/or conservation projects throughout the country or contribute to the government's support for community forestry projects.

Create a Traditional Knowledge Digital Library (TKDL) database.

Implement a tender system for bioprospecting arrangements that identifies potential genetic resources or practices of willing communities and links them to bioprospecting businesses.

Logging

The state should prohibit the clearing of any patch of forest larger than 1000 m² (0.1 ha) and of patches not surrounded by intact primary forests.

The state should prohibit the use of bulldozers and high lead yarders in logging operations.

The state should prohibit mechanical scarification of soils in forest environments. Exceptions may apply to reforestation projects of agricultural lands and roads where forest was cleared before the establishment of the West Papuan Green State.

The state should prohibit the application of chemical herbicides and pesticides in all logging operations, tree plantations, and agroforestry projects.

The state should prohibit the construction of logging roads and skid trails, as well as the use of helicopters, farm tractors, and skidders and any other form of motorized heavy machinery in forests, unless their use is explicitly planned for in the forest management plan and has received official approval by the community managing the territory as well as by the national forest authority.

The state should prohibit the practices of enrichment planting, thinning, controlled burning, vine cutting, and tree girdling by CCPs, unless their use is explicitly planned for according to RIL principles in the forest management plan, and has received official approval of the community managing the territory as well as by the national forest authority.

The state should allow for the use of enrichment planting, thinning, controlled burning, vine cutting, and tree girdling by CFEs and non-incorporated communities if they are part of traditional practices and applied selectively.

The restrictions listed here must not be enforced when logging activities solely pertain to subsistence use, nor in any community that does not engage in commercial activities.

Scenario 1: no industrial activity

Prohibit all logging for commercial activity for corporations, communities, and individuals.

Prohibit the internal trade of uncut logs, as well as the export of all wood products, high-value artisanal products and art and wood used for research purposes may be exempt.

Adopt *commercial restrictions* or *no industrial activity* scenario for land use conversion.

Scenario 2: community industry

Prohibit corporations and CCPs from engaging in logging operations.

Allow CFEs, communities without a business registration, and individuals to engage in small-scale selective logging operations.

Allow for the sale and trade of logs and wood products.

Prohibit foreign corporations from investing in logging and land governance projects.

Adopt *commercial restrictions* or *no industrial activity* scenario for land use conversion.

Scenario 3: restricted industry

Allow CCPs, CFEs, communities without a business registration, and individuals to engage in logging operations.

Allow for the sale and trade of logs and wood products.

Competing land uses

The state should prohibit large-scale (patches over 1000 m² (0.1 ha) or exceeding 4% of the territory's surface area) monocultural plantations and open-field agriculture.

The state should allow agroforestry practices that retain the existing canopy and grow multiple native crops in conjunction, but such agriculture must not cover more than 30% of a territory.

The state should only allow open field agriculture (including swidden) for commercial purposes and/or using cash-crops if they are surrounded by natural forest and returned to a fallow state within 5 years of establishment and fallow for longer than 10 years afterwards.

The state should only allow monocultured tree plantations and perennial food crops if they are community-owned, on small patches surrounded by natural forest, and returned to a forest state after cultivation.

The state should only allow cash-crop agroforestry if they are not preceded by clearing of land and canopy cover is maintained throughout transition, and if multiple tree species that are native and/or multifunctional are cultivated.

The state should promote practices that retain the existing canopy, practice intercropping, or use native plants.

Scenario 1: no industrial activity

Prohibit all forest land conversions for commercial activity for corporations, communities, and individuals.

Allow trade of cash- and food crops cultivated in forest environments on local markets only.

Complement this legislation by researching and investing in a(n inter)national traceability scheme for agricultural crops grown in existing agricultural and other non-forested areas.

Scenario 2: community industry

Prohibit corporations and CCPs from engaging in agricultural and agroforestry projects on any land that is forested as of the date of independence.

Allow CFEs, communities without a business registration, and individuals to engage in small-scale land conversions.

Allow small-scale (<0.1 ha) land conversions for cash-crop agroforestry, cash-crop plantations, and perennial food crops (i.e. orchards) using native and naturalized plants.

Allow medium-scale (<1 ha) agroforestry projects only if they use native or naturalized plants and retain the existing canopy (i.e. do not cut trees over 20m in height and retain a total canopy cover of at least 70%)

Allow trade of cash- and food crops on local and international markets.

Scenario 3: restricted industry

Allow CCPs, CFEs, communities without a business registration, and individuals to engage in small-scale land conversions.

Allow small-scale (<0.1 ha) land conversions for cash-crop agroforestry, cash-crop plantations, and perennial food crops (i.e. orchards) using native and naturalized plants.

Allow medium-scale (<1 ha) agroforestry projects including those that use exotic plants and cash-crops only if they retain the existing canopy (i.e. do not cut trees over 20m in height and retain a total canopy cover of at least 70%)

Allow trade of cash- and food crops on local and international markets.

Competing policies and government projects

The state must have the duty to safeguard traditional agricultural methods including swidden farming and home gardens.

The state must avoid policies and discourage practices promoting agricultural intensification of any sort, including the shortening of fallow times.

If the state grants subsidies for agriculture, it must consider applying these to maintained primary and secondary forests as well.

The state should avoid the subsidization of specific cash-crops and prioritize native species that grow well in community with other plants and a.) provide multiple benefits for humans such as fruit and timber (e.g. Galip, Okari, Breadfruit, Massoia), or b.) give a high return to land (e.g. Sandal- or Agarwood; hardwood like Merbau, Matoa, Kwila, Hoop or Klinkii pines).

The state should develop a new assessment tool for food security and poverty that includes at least NTFPs, nutritional quality, and health status.

The state should discourage population growth and resettlement in regions with high forest cover and low population density.

The state must restrict the construction of (logging) roads and settlements in intact forests.

The state must prioritize upgrading existing road capacity over building new roads and widening roads.

The state should employ strategic road development planning that integrates a measure of relative forest value including human and more-than-human values.

The state must collaborate with and find approval from the communities affected by infrastructure projects during the planning process.

The state should disincentivize private vehicle ownership by investing in public transport networks and in pedestrian and cycling infrastructure.

The state should promote outfitting urban areas with green spaces and permeable pavements.

The state should prohibit industrial mining projects.

The state should prohibit the creation of new pastureland.

Threatened species

Extraction

The state should embrace and promote customary practices of forest protection and resource governance such as *sasi* and *teter* to protect vulnerable species locally.

The state should offer, upon request, technical assistance to communities for aid with monitoring and determining sustainable offtake levels of nutritionally or culturally important but vulnerable species.

The state should encourage knowledge sharing and capacity building between communities, NGOs, and scientists to develop a more complete picture of landscape-level conservation needs, increase the scientific data available on species occurrence and behavior, and to find locally appropriate management interventions when necessary.

The state should prohibit the use of snare traps and shotguns in hunting.

Protected Areas

The state should assign all currently forested land as Indigenous and Community Conserved Areas (ICCAs).

In the long-term and after consultation of the affected communities, the state should pursue the conversion of all existing protected areas to ICCAs.

The state should recognize secondary forest stands and fallow stands as forests in early stages of succession rather than as degraded lands.

CITES

West Papua should become party to the CITES treaty and implement the required legislation on a national level.

The state should prohibit the trapping of and trade in any live wildlife or animal body parts for international markets. Exceptions may include extraction for research.

Carbon stocks & watersheds

The state should impose the responsibility of each person, community, and government body to protect the carbon stocks and water bodies within a territory.

The state should prohibit industrial and commercial development of plantations on peat soils.

The state should prohibit the drainage of peat soils and major earthworks in and around forests on peat soils.

The state should invest in raising awareness and finding alternative practices for peat conversion and swidden agriculture for communities living on peat soils and/or in mangroves.

The state should allow and may promote the growing and trade of Sago on peat soils under the condition that the area stays sufficiently inundated.

The state should invest in research and development of other forms of paludiculture practices by Papuan communities.

Protected Areas

The state should prohibit logging and land conversion for commercial purposes in forests within coastal and riverine buffer zones (100m from the waterside at high tide, including mangrove forests).

If hard restrictions are placed on cultivation on peat soils, High Carbon Stock areas, and/or mangrove forests, the state should compensate communities on peat soils for missed potential opportunities from commercial exploitation, possibly through a fund financed by carbon taxes from international travel.

Recommendations for further research

How a transition can be made from the current authorities to the envisioned Green State in as peaceful and safe a manner as possible.

How the government can engage in constant and meaningful dialogue with communities without incurring excessive costs, and if principles of deliberative democracy may play a role in this.

How the language and participatory mechanisms employed by the government may best accommodate the different realities of Papuan peoples, and how controlled equivocation might play a role in this.

How individuals and communities of non-Papuan descent that choose to remain in an independent West Papua can be treated as full citizens of the nation without having to suffer any kind of marginalization, and how they may be encouraged to live in accordance with any relevant customary rules and practices.

How non-forest natural environments, in particular savannahs and marine environments, may be governed, and to what extent this forest policy is applicable to those fields.

How to generate sufficient state income to cover the expenses associated with the services it envisions to provide without promoting industrial activities that risk incurring significant harm to the natural environment and Papuan cultural and economic livelihoods.

How to provide security to institutions that are important for democratic governance such as freedom of press, expression, and assembly without reinforcing colonial modes of governance.

Useful resources

Forest Politics in Indonesia: Drivers of Deforestation and Dispossession (Berenschot, et al. 2023): <https://www.forestpeoples.org/sites/default/files/documents/Forest%20Politics%20in%20Indonesia%20-%20%20Policy%20Briefing.pdf>

License to Clear: The Dark Side of Permitting in West Papua (Greenpeace International, 2021): <https://www.greenpeace.org/international/publication/47071/licence-to-clear>

Assault on the Commons: Deforestation and Denial of Forest Peoples' Rights in Indonesia (Colchester, et al. 2014): <https://www.forestpeoples.org/sites/fpp/files/publication/2014/12/assault-commons.pdf>

The Ecosystem Approach: Five Steps to Implementation (IUCN, 2005): <https://portals.iucn.org/library/efiles/documents/cem-003.pdf>

Forty Years of Community-based Forestry: a Review of Its Extent and Effectiveness – FAO Forestry Paper (Gilmour, 2016): <https://www.fao.org/3/i5415e/i5415e.pdf>

Company-community Forestry Partnerships: from Raw Deals to Mutual Gains? (Mayers and Vermeulen 2002): https://www.researchgate.net/profile/James-Mayers-2/publication/328410601_Company-community_forestry_partnerships_from_raw_deals_to_mutual_gains/links/5bcb997c299bf17a1c64397c/Company-community-forestry-partnerships-from-raw-deals-to-mutual-gains.pdf

All the Birds are Gone: Indigenous Women Speak Out Against Forest Loss in Papua (Papuan Women Group (PWG), 2021): https://asia-ajar.org/wp-content/uploads/2021/03/ABAG_FINAL_compressed.pdf

Additional resources:

Nesting, Subsidiarity, and Community-based Environmental Governance beyond the Local Level (Marshall, 2008): <https://www.jstor.org/stable/26522991>

Forest Management and Conservation of Biodiversity: An Overview (Putz, et al., 2001): <https://www.jstor.org/stable/2641641>

Harvesting Non-timber Forest Products Sustainably: Opportunities and Challenges (Ticktin & Shackleton, 2011): https://doi.org/10.1007/978-3-642-17983-9_7

A Classification of Threats to Traditional Ecological Knowledge and Conservation Responses (Tang & Gavin, 2016): <https://www.jstor.org/stable/26393228>

Biodiversity conservation: How can the regulation of bioprospecting under the Nagoya Protocol make a difference? (Chiarolla et al, 2013): https://www.cifor.org/publications/pdf_files/articles/APirard1301.pdf

The Complexities of Community based Tourism: how External Forces Influence the Sustainability of Bird Watching Tourism in the Remote Arfak Mountains of West Papua (Everard, et al., 2019): [https://www.researchgate.net/profile/Sharon-Harwood/publication/315455869_The_complexities_of_community_based_tourism_how_external_forces_influence_teh_sustainability_of_bird_watching_tourism_in_the_remote_Arfak_Mountains_of_West_Papua/links/58d0aa4992851c8841c28aa1/The-complexities-of-community-based-tourism-how-external-forces-influence-teh-sustainability-of-bird-watching-tourism-in-the-remote-Arfak-Mountains-of-West-Papua.pdf](https://www.researchgate.net/profile/Sharon-Harwood/publication/315455869_The_complexities_of_community_based_tourism_how_external_forces_influence_the_sustainability_of_bird_watching_tourism_in_the_remote_Arfak_Mountains_of_West_Papua/links/58d0aa4992851c8841c28aa1/The-complexities-of-community-based-tourism-how-external-forces-influence-teh-sustainability-of-bird-watching-tourism-in-the-remote-Arfak-Mountains-of-West-Papua.pdf)

The Current Legal and Institutional Framework of the Forest Sector in Papua New Guinea (Neil, et al., 2007): https://cdn.odi.org/media/documents/92_QmbrON7.pdf

Revisiting the Issue of Elite Capture of Participatory Initiatives (Lund & Saito-Jensen, 2013): <https://www.sciencedirect.com/science/article/pii/S0305750X1300034X>

Avoiding Legality: Timber Producers' Strategies and Motivations under FLEGT in Ghana and Indonesia (Acheampong & Maryudi, 2020): <https://www.sciencedirect.com/science/article/pii/S1389934118304441>

FLEGT Regulation: Union-Wide Overview for the Year 2021 (European Commission, 2021): <https://agrikoltura.gov.mt/en/phd/Documents/reportObligations/flegtOverview2021.pdf>

Traceability Systems: Potential Tools to Deter Illegality and Corruption in the Timber and Fish Sectors? (Grant, et al. 2021): <https://bridgebiointegration.net/current-global-projects/tnrc/resources/topic-brief-traceability-systems-potential-tools-to-deter-illegality-and-corruption-in-the-timber-and-fish-sectors.pdf/view>

Annexes

Annex 1: Glossary

ABS: Access and Benefit Sharing – a set of principles which should be followed for the fair and just access to and commodification of natural resources as formulated by the Convention on Biological Diversity

CBF: Community-based Forestry – any program that involves local communities in the planning and/or daily management of commercial forest activities. The degree of community involvement may range from simple consent-seeking procedures to full devolvement of all relevant rights and responsibilities.

CBNRM: Community-based Natural Resource Management – any program that involves local communities in the planning and management of any or all relevant natural resource(s).

CCP: Corporate-Community Partnership – a formal collaborative agreement between a corporation and a defined community that is established for the commercial exploitation of a product, place, or practice.

CFE: Community Forestry Enterprise – a community-owned and -led business entity that is established for the commercial exploitation of a product, place, or practice present in forests.

CFM: Community Forest Management – any program that involves local communities in the planning and management of forest activities.

GSV: Green State Vision

ICCA: Indigenous & Community Conserved Area – a land area dedicated to the conservation of species richness and ecosystem functions under the management of a community of Indigenous or local people.

LUCC: Land Use and Cover Change – processes that alter the primary use of land so significantly that it greatly affects the composition of life and functioning of biochemical cycles upon the land. In the context of this report LUCC refers to the change from Intact Forest Landscapes with a high degree of tree cover and only low-impact

subsistence use of forest resources to any other form of land use.

MAT: Mutually Agreed Terms – an agreement specifying the conditions of access to specific resources and benefit-sharing between two parties (e.g. a company and a local community).

MBI: Market-based Instrument – any economic program that seeks to achieve a non-economic goal (i.e. forest and nature conservation) by pursuing funding and poverty reduction through free-market mechanisms.

NTFP: Non-Timber Forest Product – any material extracted from forest environments that is not timber, including those derived from flora, fauna, fungi, and soil; used for subsistence or commercial purposes.

PA: Protected Area – a land area dedicated to the conservation of species richness and ecosystem functions by means of restricting forest access and regulating forest activities.

(F)PIC: (Free) Prior Informed Consent – the UN-recognized right for peoples to be able to decide freely whether to reject or allow any project that affects them or their territories.

RIL: Reduced Impact Logging – a set of guidelines (pre-harvest inventories; harvest intensity; cutting cycles; skid trail planning; liana cutting) that aims to minimize the residual damage of timber extraction.

SRA: Social Responsibility Agreement – a formal agreement specifying which responsibilities and social obligations a company has towards the community that governs the territory where it operates.

TEK: Traditional Ecological Knowledge – ways of life that are formed and informed by an intimate knowledge of and relationship with a specific natural environment, and which have been transmitted in-place across multiple human generations.

Bioprospecting – activities relating to the search for species and genetic resources from which commercially valuable compounds such as medicinal drugs can be derived.

Commons and common pool resources – a system of openly accessible renewable resources that has limited capacity to regenerate e.g. a forest or an animal population; their governance requires shared rule formulation and compliance.

Ecotourism – tourism in which the main commodity is ecosystem-based and which aims to conserve the relevant ecosystem; includes wildlife watching and cultural activities.

Human-nature and nature-culture dualism – the idea and mode of thinking that regards human and human society as separate from nature and natural processes; is conducive to naturalizing the subordination of non-human beings and to justifying resource extraction and management; is incompatible with the concept of guardianship.

Intact Forest Landscape – a continuous mosaic of forest, consisting of tree stands of different ages and non-forest patches, large enough to maintain all native biological diversity including viable populations of wide-ranging species.

Keystone species – species that are particularly important in the creation, transformation, and persistence of habitat conditions; without these species their ecosystems would degrade and disappear over time.

Land tenure – the relationship between people and land as determined by legal and customary regimes. Specifically it refers to the holding of land by specific groups or individuals, which determines who decides on the use of land and who has access to benefits derived from its use.

Natural gap dynamics – natural forests are in a state of dynamic equilibrium, where various disturbances (incl. tree falls, fires, etc.) cause gaps to emerge in the forest canopy that change local habitat conditions and allow different species to thrive; if the frequency and intensity of disturbances are proportional to the forest's size, they produce a highly biodiverse landscape.

Payment for Ecosystem Services (PES) – any program that aims to offer the guardians of an ecosystem payments for the continued provision of the services that the ecosystem provides; an MBI that is supposed to be paid for by the users of the relevant ecosystem service but in reality often remains reliant on government subsidies.

Political economy and ecology – (the study of) how political and economic aspects of society interact, with particular regard to environmental changes. Focus points in the context of forests include access to land and resources and structural (money; access to politicians and bureaucracy) and discursive (storytelling, defining success) power asymmetries in decision-making.

Polycentricity – a system of governance that unites multiple relatively autonomous decision-making centers through an overarching system of rules and goals and the exchange of spontaneously emerging ideas and institutions.

Reducing Emissions from Deforestation and Forest Degradation (REDD+) – a PES framework established by the UNFCCC that aims to provide payments to landowners for retaining forest cover relative to a baseline deforestation rate in order to mitigate climate change.

Sasi – a form of environmental governance and Papuan customary law that restricts or imposes temporary bans on harvesting activities in given areas, at certain times, or of specific species.

Swidden and fallow agriculture – a form of shifting agriculture that relies on slash-and-burn (swidden) practices before crop cultivation and abandonment (fallow) after; this important element of traditional agriculture and human livelihoods in tropical forests is often regarded as harmful and a driver of deforestation but is compatible with sustainable forest use if fallow times are sufficiently long.

Tragedy of the commons – the idea that common pool resources will inevitably degrade under the influence of population growth because each user prioritizes their self-interest over the common interest, which leads to overextraction.

Annex 2: Guiding principles for a Forest Policy Brief in line with the GSV

Forest policy in West Papua should:

- Safeguard balance and harmony amongst human and non-human beings, protecting the right to life of each organism and species, which means to:
 - o Provide humans with sufficient access to means to provide in their livelihoods.
 - o Maintain species richness and prevent extinction and exploitation of non-human beings, which requires policies to:
 - Ensure the maintenance of habitats with sufficient food, shelter, and rest for all species.
 - Maintain environmental quality and health and ensure habitat connectivity.
 - o Ensure the role of humans as guardians, protectors, and carers of non-human beings.
- Prevent the further emission of GHGs locally and globally and, where possible, promote carbon capture and climate resilience, which requires policies to:
 - o Not support or enable emissions elsewhere.
- Recognize and criminalize ecocide.
- Govern land use and cover change from forests to use by other industries, under the assumption they are employing best environmental practices, which requires policy to:
 - o Consider and weigh the tradeoffs of land use changes in terms of their environmental, climate, biodiversity, and livelihood impacts.
- Where possible, allow for the generation of state income for the support of social protection programs.
- Recognize, respect, and maintain the Traditional Ecological Knowledge and customary norms and values of Indigenous and local communities, which means:
 - o Customary authorities and practices should be leading in developing forest governance approaches, provided they can safeguard the social and environmental objectives of the GSV.
- Provide a legal framework to support effective decision-making and guardianship of forests.
- Ensure that generated benefits from forests flow towards both local communities and regional/national social protection and welfare programs.
- Ensure the protection of forests and their environmental conditions from abuse by customary authorities.
- Promote equitable access to forests and their derived benefits for women, children, and racial, ethnic, tribal, and religious minorities.
- Involve key stakeholders in all stages of decision-making regarding forests that affect their rights and interests.

Annex 3: Consulted persons

Description	Topic(s) of discussion	Type of contact
Female Indigenous Papuan church leader and human rights worker.	Women in decision-making; customary governance; state-community relations	Multiple videocalls
Male Indigenous Papuan volunteer for ENGO and forest campaign advocate.	Community-corporate relations; forest policy	Phone call; videocall; text messages
Papuan diaspora conservation student.	Forest policy; community-based conservation	In person conversations; text messages
Dutch ecological researcher and environmental activist.	Community-corporate relations; resource extraction	In person conversations; text messages
Indonesian government employee & environmental governance researcher.	Polycentricity; state-community relations	Email contact
Papuan diaspora human rights lawyer.	Women in decision-making; rights and legal system	Videocall; brief revision
Dr. Sophie Chao – author; environmental anthropologist and humanities scholar	Recognition of customary territories; Papuan ontologies	Videocall; brief revision
Various panel speakers in conference on ecocide in West Papua.	GSV; decolonial governance; state-community relations	Open panel discussion

Annex 4: Multifactor Analysis of Silvicultural Practices

Logging methods

	Livelihoods	Biodiversity	Ecosystem functions	Carbon stock	Associated risks	Productivity	Investment required
Clear cutting/ conventional logging	Highly profitable, erasure of subsistence foraging/ ²⁰⁸ hunting ²⁰⁸	Severe effects on flora and fauna (genetic, species, community & ecosystem level), reduction of forest specialists, increased invasives and pioneers ²⁰⁸ , 22% species reduction (bird, fungal, arthropod) ²⁰⁸ , fragments arboreal fauna ²⁰⁹	Reduced uptake and increased runoff of water and nutrients; soil erosion ²⁰⁸ , increased droughts, floods, and temperatures ¹⁷⁶	Severely reduced ²⁰⁸	Risk of arrested succession ²⁰⁸ Reduced water quality; Increased erosion on skid trails and roads ¹⁷⁶	Limited/ arrested recovery of stocks ²¹⁰	Low ²⁰⁸
Conventional Selective logging	Profitable, reduction of foraging/ ²⁰⁸ hunting ²⁰⁸ , negative effect on NTFPs ¹¹²	Reduction of genetic and species diversity; community diversity preserved if combined with reserves ²⁰⁸ , No overall species richness effect ²⁰⁸ , Limited effect on genetic diversity ²¹¹ ; mangrove species richness sustained ²¹³	Increased runoff, no flooding ¹⁷⁶ , Limited effect on soil erosion ²⁰⁹	Limited reduction ^{208,177} 76% retained, 100% recovery after 16 years ¹⁷⁶	Limited erosion on skid trails and roads ¹⁷⁶	Medium-high ²⁸ , reduced collateral damage aids restocking rate ¹⁷⁷	High: detailed inventory & planning and worker education ²⁰⁸
Reduced impact logging	Profitable, reduction of foraging/ ²⁰⁸ hunting ²⁰⁸ ; negative ¹¹² or no ²¹² effect on NTFPs	No overall species richness effect, fungal and arthropod diversity reduced ²⁰⁸	Limited effect on soil erosion ¹⁷⁹	Need harmful interventions ²⁰⁹ ; Limited road use and erosion ¹⁷⁹	Increased erosion on skid trails and roads ¹⁷⁶	Increased recovery stocks ¹⁷⁹ , ST: Increased stocks for some species ²⁰⁸	Low ²⁰⁸
Retention incl. Seed-tree (ST) & shelterwood (SH)	Profitable, reduction of foraging/ ²⁰⁸ hunting ²⁰⁸	No overall species richness effect ²⁰⁸ , Limited effect on genetic diversity ²¹¹	Increased runoff, no flooding ¹⁷⁶ , No effect on sedimentation in mangroves ¹¹³	In mangroves, soil unaffected 70–75% of total retained, recovered in 15–25 yrs ²¹³	Increased erosion on skid trails and roads ¹⁷⁶		Medium ²⁰⁸
Selection incl single (SI) and group (GR)	Profitable, reduction of foraging/ ²⁰⁸ hunting ²⁰⁸	Locally severe effects ²⁰⁸ , fragments arboreal fauna ²⁰⁹					Medium ²⁸
Patch cut							

	Biodiversity loss	Carbon emissions	Associated risks	Logging speed	Productivity	Investment required
Manual	Low ^{28,209}	None	None ²⁸	Low		None ²⁸
Draft animal	Low ^{28,180}	Low	None ²⁸	Low-medium		Low ^{28,180}
Farm tractor	Medium ²⁸ , medium soil damage ¹⁸⁰	Medium		Medium	Reduced due to soil damage ²⁸	Low-Medium ²⁸
Skidder	Medium ²⁸ , medium-high soil damage ¹⁸⁰	Medium		Medium	Reduced due to soil damage ²⁸	Medium ²⁸
Bulldozer	Excessive damage to residual trees and soil ^{28,180}	High	Road construction ²⁸	Medium-High	Severely reduced due to soil & residual tree damage ^{28,180}	High ²⁸
Balloon	Low ²⁸	Low	Expansion into difficult terrain ²⁸	Low-medium		
Skyline	Low-Medium direct impact ^{28,209}	Medium	Road construction; Expansion into difficult terrain ²⁸	Very high		Very High ²⁸
High lead	Excessive ²⁸	Medium	Road construction; Expansion into difficult terrain ²⁸	High	Severely reduced due to residual tree damage ²⁸	High ²⁸
Helicopter	Low direct impact ^{28,209}	High	Road construction; Expansion into difficult terrain ²⁸	High		Very High ²⁸

Yarding methods

	Livelihood	Biodiversity	Ecosystem functions	Carbon Stock	Productivity	Investment & Labor	Notes
Controlled burning		Increased risk of invasiveness ²¹⁴	Reduces necromass/ carbon stocks ²⁸ , increased water retention ¹⁷⁸	Increased emissions ²⁸	Increased ²¹⁴		Reduces fire risk ¹⁷⁸
Mechanical scarification		Reduced microbe functional ²¹⁵ and nematode ²¹⁴ diversity in soils; mixed/temporary effect on flora; reduces fauna ²¹⁴	Disturbs soil hydrology & fertility, increased erosion & nutrient loss ^{28,214} ; Loss microbial function ²¹⁵ ; Reduced water quality ²¹⁴	Initial decline, long-term increased capture ²¹⁴	Increased initial growth; long term unknown ²¹⁴	Medium: raising and planting seedlings ²¹⁴	Mostly relevant for reforestation or recovery after soil compaction ²¹⁴
Herbicides		Severe reduction bird diversity ²¹⁶	Decreased plant germination, growth & photosynthesis ¹⁷⁸ ; Increased erosion and runoff ²¹⁷ ; reduced water quality ²¹⁸	Limited decline ²¹⁸	Increased ^{214,219} ; risks arrested recovery ¹⁷⁸	Low ²¹⁴ – medium ²¹⁹	
Enrichment planting	Possible revenue from REDD+ and NTFPs ¹⁷⁷	Severe reduction at all levels of diversity ²⁸ , limited impact on birds & invertebrates ¹⁷⁷	Reduced climate resilience ¹⁷⁷	Increased capture ¹⁷⁷	Severely increased, low-high profitability ²⁸ Increased recovery ^{177,220}	High: other treatments required ²⁰⁹ ; raising and planting seedlings ¹⁷⁷	Depends on species and planting density ²⁰⁹
(Liberation) thinning	Depletes NTFPs ¹¹²	Reduces floral & faunal species & genetic diversity ^{28,177} ; fragments arboreal fauna ²⁰⁹	Pollination, seed dispersal, carbon sequestration & predation altered ²²¹ Aboveground biomass increased; reduced climate resilience ¹⁷⁷	Decreased capture ²²¹ ; Increased capture ¹⁷⁷ ; initial decline, long-term increased capture ²¹⁰	Increased ^{28,221} ; 9-27% increase if combined with vine cutting ²²²	Low ¹⁷⁷	
Vine cutting	Depletes NTFPs ¹¹² ; Promotes NTFPs if selective ¹⁷⁷	Negatively affects flora and fauna ^{28,177} ; fragments arboreal fauna ²⁰⁹	Increases aboveground biomass ¹⁷⁷	Increased capture ¹⁷⁷	Increased ²⁸ ; 9-27% increase if combined w/ thinning ²²²	Low ¹⁷⁷	
Girdling competing trees		Erases floral diversity and reduces genetic diversity ^{28,223}	Severely reduces biomass/carbon ²⁸ ; reduced climate resilience ¹⁷⁷	Increased emissions ²⁸	Increased ^{220,223}	Low ¹⁷⁷	

Management interventions^v^v Impacts depend on intensity of the intervention's application. Interventions are best applied selectively to crop trees.

Annex 5: Impacts of Land Use and Cover Change (LUCC)^{vi}

1/2	Livelihoods	Economic opportunity	Biodiversity loss	Ecosystem functions	Carbon ^{vii}	Climate resilience	Suitable crops / additional notes
Oil palm plantation	Negative ²²⁴ , Harms human health ^{235,232} ; Mixed for migrant workers ¹³⁹	Lower than value of forest ecosystem services ²²⁴ ; High ²²⁶ , High from employment ²²⁷	54% ²⁰⁸ very severe ^{viii} for birds, only common species conserved ²²⁸	Decreased soil stability, increased erosion, cause of coral mortality ²⁰⁷ ; severe reduction of most functions, increased plant pests, water & air pollution ^{200,229,229}	BGC -13% ²³⁰ ; AGC -65%; Excessive CE from peat fires and drainage ¹⁹⁶ ; TEC medium ²³¹ ; TEC -61% ²³² to -62% ²³³	Low ^{234,235}	Oil palm
Non-timber plantation (rubber)	Similar to oil palm, better return to land, less to labour, diversified income ²³⁶	High ²²⁶ , Medium from employment ²²⁷	54% ²⁰⁸ , severe for plants, animals, and soil fauna & microbes ²³⁷ ; severe for birds, common species conserved ²²⁸	Decreased ²³⁷	BGC -13% ²³⁰ , TEC medium-high ²³¹ ; TEC -56% ²³²	Low ²³⁵	Rubber tree
Timber plantation	Severe loss of culture & wild food sources ^{238,239}	Medium ^{208,226} , Increased income from timber ^{238,240}	40% ²⁰⁸ , severe ²³⁸ , severe for birds, common species conserved ²²⁸	Reduction of all ecosystem functions, decreased water quality ²³⁸ , decreased soil health ²⁴¹	BGC -13% ²³⁰ , TEC medium-high ²³¹ ; reduced ²²⁸	Low drought resistance ²⁴²	Native: Kwila; hoop pines; klinkil pines ²⁴³ , Lawang; Merbau; Matoa ²⁴⁴ ; Exotic: Mahogany; Damar; Rosewood; Teak ^{240,243}
Pulp- and fuelwood plantation	Severe loss ^{239,245} , Perceived worse than timber ²⁴⁶	Low ^{208,245,246} , diversification under specific conditions ²⁴⁵	40% ²⁰⁸ , Perceived worse than timber ²⁴⁶	Perceived worse than timber ²⁴⁶	BGC -13% ²³⁰ , TEC medium-high ²³¹		Acacia ²⁰⁸ ; Eucalyptus ²⁴⁰
Aromatic wood plantation	Little cultural relevance, improved income ⁸⁴	Very high, but difficult to achieve ²⁴⁷	Similar to other tree monocultures	Similar to other tree monocultures	BGC -13% ²³⁰ , Low ²⁴⁸		Sandaiwood; Eagle- or Agarwood ²⁴³ ; Benzoin; Caiuput ²⁴⁹ ;
Native food & timber agroforestry	Improved ^{250,251}	Big income increase ^{244,252} , reduced timber productivity ²⁴¹	None – low ^{250,251,253} , limited for birds, rare species conserved ²²⁸	Limited soil erosion and nutrient runoff ²⁵⁴ , maintained soil processes ²⁵⁵	TEC medium ²³¹ ; TEC – 51% ²³³	Decent drought & flood resistance ^{242,256}	Galip; okari; coconut; breadfruit; neen ²⁴³ ; Rattan ²³⁶ ; Lawang; merbau; durian; matoa; rambutan; cinnamon; massolia; nutmeg ^{244,257}
Cash-crop agroforestry	Improved ¹⁸²	Better than monoculture due to input reduction & increased resilience ⁶⁷ ; Big income increase ^{244,252} ; Low for oil palm ²⁵⁸ ; High for rubber ²⁵⁹ ; Increased productivity ²⁶⁰	32% for exotic, non-timber species ²⁰⁸ , limited reduction for rubber ²⁵⁷ ; functional bird diversity loss at community level ²⁶¹	Limited soil erosion and nutrient runoff ²⁵⁴ , limited reduction for rubber ²⁵⁷ ; maintained soil processes ²⁵⁵	TEC medium ²³¹ ; TEC – 51% ²³³ ; TEC -41% for rubber	Decent drought & flood resistance ^{242,256}	Rattan ²³⁶ ; Avocado ²⁵⁷ ; Rubber tree; coffee; resin trees; oil palm; cacao; pepper ^{182,236,262} ; Teak; sandaiwood; eaglewood; mahogany; rosewood ²⁴³

^{vi} Relative to used and selectively logged tropical forests, continued on next page.

^{vii} AGC = Above ground carbon, BGC = Below ground carbon, TEC= total ecosystem carbon, SOC= Soil Organic Carbon, CE= Carbon emissions.

^{viii} To indicate the impact of changes from least impact to most impact we use the following scale: None – Limited – Medium – Significant – Severe.

2/2	Livelihoods	Economic opportunity	Biodiversity loss	Ecosystem functions	Carbon ^{vi}	Climate resilience	Suitable crops / additional notes
Pasture		Low ²⁶³	Medium-high ²⁵³	Decreased soil processes and health ^{255,264}	BGC 0 to +8% ²³⁰ ; TEC low-medium ²³¹ ; SOC -31% ²⁶⁵	Medium, increased pests, fire & floods ^{266,267}	Maiz; Sweet potato; cassava; taro; caladium; mustard green; water spinach; gedi; peanuts ²⁶⁷
Annual food crops	Significant decline ²⁶⁸	Initial increase long-term loss ²⁵¹	High ^{251,253} ; functional bird diversity loss at community level ²⁶¹	Significant decline in soil health ²⁶⁶ ; significant decline ²⁵¹	BGC -42 to -53% ²³⁰ ; TEC low ²³¹ ; rice paddy TEC -73% ²³³ ; AGC+SOC severe decline ²⁶⁸	Low ²⁶⁶	Maiz; Sweet potato; cassava; taro; caladium; mustard green; water spinach; gedi; peanuts ²⁶⁷
Perennial food crops	Medium decline ²⁶⁸ ; mixed ²⁶⁹	Potential for diversification & increase ^{268,269}	Medium-high ²⁵³ ; functional bird diversity loss at community level ²⁶¹ ; limited ²⁶⁹	Significant decline in soil health ²⁶⁶ ; limited decline ²⁶⁹	TEC medium-high ²³¹ ; AGC + SOC severe decline ²⁶⁸		Chilli; Banana; Ginger; Turmeric; Cardamom; Avocado; Durian; Rambutan; Breadfruit; Coconut; Coffee; Cacao; Breadfruit ²⁶⁷
Swidden long-fallow (>10 yrs)	None to slightly positive ²⁶⁸	Potential for diversification & increase ²⁶⁸	None – low ^{251,270}	None – limited decline ²⁵¹	High initial EC but high recapture ²⁶⁸		
Swidden short fallow (<5 yrs)	Significant decline ²⁶⁸	Negative ²⁷⁰	Significant ²⁷⁰	Significant decline in soil health ^{251,268}	AGC + SOC significant decline; high EC; limited recapture ²⁶⁸		
Mining	Severe decline for forest and fishing; severe health effects ²⁷¹	Very high ²⁷² , lower than biodiversity & eco-tourism ²⁷³	Severe, especially for copper & nickel, incl. outside of mine area ^{271,274}	Severe & long-lasting decline, incl. water pollution and scarcity, cause of coral mortality ²⁷⁵	TEC removed, EC high ^{271,275}	Low resilience, high risk ²⁷²	Lots of indirect land use change ²⁷⁵
Road construction	Negative health effects ²⁷¹ , improved, increase market-based orientation ²⁷⁶	High ²⁷⁶ ; on global level relatively low in Papua ^{277,278}	High ²⁷⁶ ; Devastating due to fragmentation and increased hunting and logging access ^{28,276}	Severe decline of soil health and water quality ²⁸	Loss of TEC; EC high and increased EC from traffic ²⁷⁹		Lots of indirect land use change ^{9,28,30,280}
Urban development	Improved, increased market-based orientation, decreased health ^{281,282} , decline in fishing and culture ²⁸³	Mixed, increased inequality ^{282,284}	Severe decrease of fish abundance & mangroves ^{283,285}	Decreased water quality and wild food provision ²⁸³ ; Increased flooding, air pollution, and disease ²⁸²	Severe increase in EC ²⁸¹		Severe indirect land use change ^{109,264,282}

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